

CHEMOPD: MULTI-TEACHER ON-POLICY DISTILLATION FOR MULTI-TASK CHEMICAL REASONING

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ABSTRACT

Large language models are increasingly expected to support diverse chemical reasoning capabilities within a unified model. One approach is to develop specialized capabilities separately and consolidate them through multi-teacher on-policy distillation, but this raises two questions: how should specialization be organized, and how should specialist guidance be integrated? We introduce ChemOPD, which addresses both. We estimate task affinities from supervised fine-tuning gradients and solve a constrained mixed-integer program (MIP) to construct partially overlapping specialist groups. During distillation, we retain a generalist teacher trained on all tasks so that specialist guidance supplements rather than replaces its supervision. Our anchor-residual objective gradually increases the routed specialist’s contribution on student-generated responses. On ChemCoTBench, affinity-guided specialization produces task-dependent gains over the generalist teacher and improves several capabilities beyond semantic task grouping. Yet stronger teacher-side performance does not automatically yield stronger students: with the same specialists and routes, anchor-residual OPD improves most reported metrics over specialist-only distillation and realizes a larger share of the available teacher gains. These results highlight specialization and capability integration as connected but distinct design problems in chemical reasoning.

1 INTRODUCTION

Large language models (LLMs) are increasingly being developed to solve a broad range of chemical problems. Beyond chemical question answering, recent models have been applied to molecular property prediction, molecule generation and editing, reaction prediction, retrosynthesis, and multi-step chemical reasoning (Yu et al., 2024; Zhao et al., 2025b; Lin et al., 2025; Zhao et al., 2026; Liu et al., 2026; Hao et al., 2026; Song et al., 2026; Wang et al., 2026b). These tasks rely on overlapping chemical knowledge, but demand different forms of reasoning and different outputs: predicting a property, modifying a molecular structure, or inferring a reaction outcome does not require exactly the same capability. As chemical language models become more general, an important challenge is therefore not only to improve individual tasks, but to develop a broad collection of chemical capabilities within a single model.

A single model at inference does not require every capability to be developed through the same training process. Knowledge distillation provides a natural way to separate capability development from capability deployment: stronger or more specialized models can acquire useful behaviors during training, and their knowledge can later be transferred into a unified student (Hinton et al., 2015; Xu et al., 2024). This idea is particularly appealing when different tasks benefit from different forms of specialization. Multi-task learning has long shown that sharing data and parameters across related tasks can be beneficial, while task-grouping studies further demonstrate that which tasks are learned together can strongly affect the capabilities that emerge (Standley et al., 2020; Fifty et al., 2021; Song et al., 2022). Thus, developing specialists is not simply a matter of assigning one model to each task: it also requires deciding which tasks should contribute jointly to each specialist.

Once specialized capabilities have been developed, they must still be made useful to a shared student. For autoregressive models, on-policy distillation (OPD) lets a teacher supervise the contexts that the student actually visits during generation, rather than only imitating completed teacher solutions (Agarwal et al., 2024; Gu et al., 2024; Song & Zheng, 2026; Li et al., 2026; Xu et al., 2026). Multi-teacher OPD further enables capabilities developed in different teachers to be consolidated into one student (Ma et al., 2026). This provides a natural route from specialization to a unified model, but leaves two choices unresolved for heterogeneous chemical reasoning: *how should specialization itself be organized, and how should the resulting specialist guidance be combined when training a shared student?*

The first question concerns how chemical tasks should be grouped when developing specialists. Human-readable task categories describe semantic differences, but they need not fully reflect how training one task affects the learning of another. We therefore ground specialist construction in measured training relationships. Using supervised fine-tuning (SFT) gradients, we estimate local task affinities (Fifty et al., 2021). Building on optimization-based task grouping (Kang et al., 2011; Wang et al., 2024), we adapt a constrained mixed-integer formulation to construct a small collection of partially overlapping specialists. The overlap is intentional: a task can contribute useful training signal to a specialist even when that specialist is not ultimately responsible for supervising the task during distillation. In this way, task relationships inform how expertise is developed without requiring benchmark categories to serve as fixed training boundaries.

The second question concerns how specialist expertise should enter the student’s learning process. Rather than rely only on routed specialists, we retain a generalist teacher trained on all tasks so that task-specific guidance supplements its supervision. Our anchor-residual OPD objective uses the generalist’s guidance as the anchor and adds a correction from the routed specialist on the same student-generated prefixes. We gradually increase the specialist weight through a linear warm-up. At inference, only the unified student is used.

We evaluate ChemOPD on ChemCoTBench (Hao et al., 2026), separating the quality of the constructed teachers from the student’s ability to benefit from them. Affinity-guided grouping produces task-dependent specialist advantages over the generalist teacher and improves several capabilities relative to benchmark-family grouping. However, stronger teacher-side performance does not automatically translate into stronger students: positive specialist headroom can coexist with poor or even negative student gains under specialist-only OPD. With the same specialist checkpoints and task routes, anchor-residual OPD realizes a broader fraction of the available teacher gains and improves most reported metrics over specialist-only distillation. Component ablations further support combining generalist supervision with gradual specialist weighting, rather than using specialists alone or applying the final mixture throughout training. Together, these results suggest that developing specialized capabilities and making them usable by a unified student are connected but distinct design problems.

Our contributions are:

- We develop affinity-guided specialist construction for heterogeneous chemical reasoning, using measured task interactions and constrained grouping to organize partially shared specialist training.
- We introduce anchor-residual on-policy distillation, which treats routed specialist guidance as a correction to a generalist teacher’s supervision and gradually increases its contribution during training.
- We separate teacher-side capability from realized student gains and show that effective capability integration depends on both the source and the training-time balance of shared and specialist supervision.

2 RELATED WORK

Chemical reasoning with language models. Large language models have been adapted across chemical prediction, molecular generation, and reasoning. Fine-tuned LLMs have been studied for molecular and materials properties, reaction yields, and inverse design (Jablonka et al., 2024), while MolT5 connects molecular structures with natural language through molecule captioning and

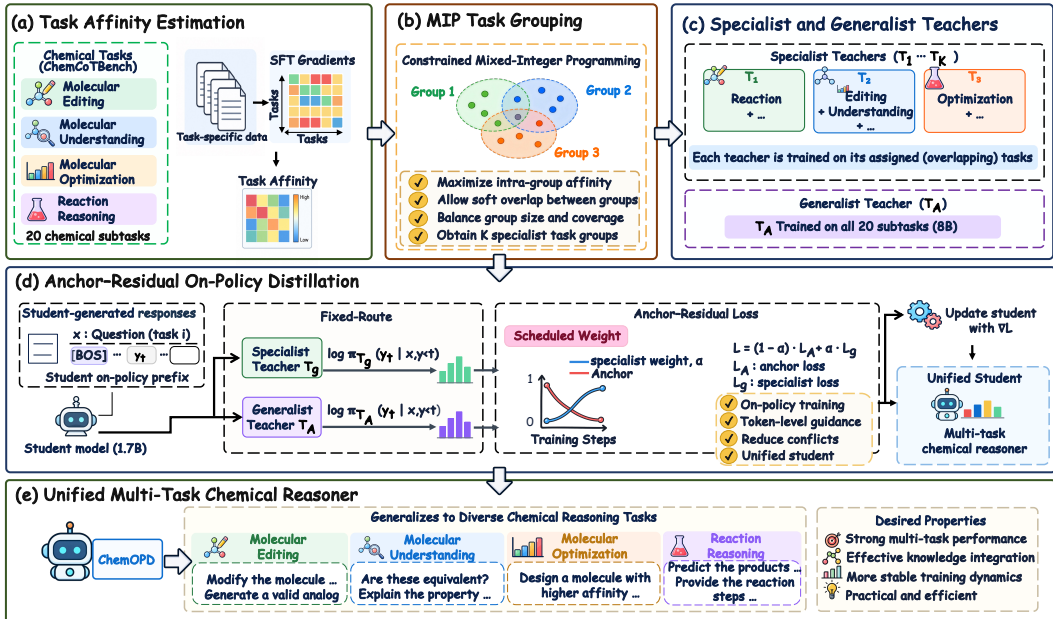


Figure 1: Overview of affinity-guided teacher construction and anchor-residual multi-teacher OPD. The MIP converts gradient-based task affinities into overlapping task groups for specialist training. During OPD, the generalist teacher and the fixed-route specialist score the same student-generated prefixes. The generalist’s advantage serves as the anchor, and the specialist residual is introduced gradually to train a unified student.

text-conditioned generation (Edwards et al., 2022). Large-scale instruction-tuning efforts such as Mol-Instructions and LlaSMol further cover diverse molecular, biomolecular, and reaction-related tasks (Fang et al., 2024; Yu et al., 2024). Chemistry-specialized models including ChemLLM and ChemDFM pursue broader domain capabilities within a single language model (Zhang et al., 2024; Zhao et al., 2025a), while ether0 explicitly studies reasoning-oriented post-training across diverse chemistry problems (Narayanan et al., 2026). ChemCoTBench complements these model-development efforts with structured evaluation of molecular understanding, editing, optimization, and reaction reasoning (Hao et al., 2026). Our work focuses on how such heterogeneous chemical capabilities can be developed through specialization and subsequently integrated into a unified student.

Task grouping in multi-task learning. Multi-task learning has long studied which tasks should share training. Prior work considers partially overlapping task groups (Kumar & Daume III, 2012), searches for effective task combinations (Standley et al., 2020), and estimates task affinity from cross-task training effects (Fifty et al., 2021). Constrained optimization has also been used to form task groups under coverage and resource constraints (Wang et al., 2024). We build on this line of work to organize the training mixtures of specialist teachers before distillation.

On-policy and multi-teacher distillation. Knowledge distillation transfers teacher behavior to a student (Hinton et al., 2015). For language models, MiniLLM and GKD study distillation with reverse-KL and student-generated sequences (Gu et al., 2024; Agarwal et al., 2024). MOPD extends on-policy distillation to multiple domain-specific teachers that are consolidated into a unified student (Ma et al., 2026), with recent work exploring other forms of multi-teacher supervision (Wang et al., 2026a; Chen et al., 2026; Zhou et al., 2026).

3 PROBLEM SETTING

Let $\mathcal{T} = \{1, \dots, n\}$ denote a collection of chemical subtasks and let D_i be the SFT data for task i . We seek m teacher groups $\mathcal{G} = \{G_1, \dots, G_m\}$. Every task must appear in at least one group; group-

size and membership constraints limit overlap. Training a teacher on $\cup_{i \in G_g} D_i$ yields specialist T_g , while a generalist teacher T_A is trained on $\cup_i D_i$. The student π_{θ_0} is initialized by all-task SFT. Each OPD prompt x has a known subtask label $c(x)$, and a routing function $r(c(x)) \in \{1, \dots, m\}$ selects one specialist. Although teacher training groups may overlap, each routed example receives one specialist route, fixed before OPD. The generalist-only baseline has no specialist route. We evaluate both whether the groups produce useful specialist advantages and whether these advantages can be incorporated into a student with broad cross-task capability.

4 METHOD

ChemOPD addresses teacher construction and capability integration in two stages. Stage I uses SFT-gradient affinities and a constrained MIP to select partially overlapping training groups for specialist teachers. Stage II combines the generalist teacher’s supervision with a residual correction from the routed specialist on student-generated responses. A linear warm-up gradually increases the specialist contribution. Figure 1 summarizes the two stages.

4.1 GRADIENT PROBES AND TASK AFFINITY

To construct specialists, we estimate whether training on one task could locally benefit another (Fifty et al., 2021). Each subtask supplies two disjoint probe sets drawn from its SFT training data: a source-update probe and a target-loss probe, referred to as the training and validation probes below.

Let ϕ denote the model parameters and ϕ_0 the teacher’s initial checkpoint. For task i , let $L_i^{\text{tr}}(\phi)$ and $L_i^{\text{val}}(\phi)$ denote the SFT losses on its training and validation probes, respectively. Each loss averages cross-entropy over valid assistant tokens within each example, then equally across examples. For a source task i and a target task j , let $\bar{g}_i^{\text{tr}}$ and $\bar{g}_j^{\text{val}}$ denote the training-probe gradient of L_i^{tr} and the validation-probe gradient of L_j^{val} . Both are evaluated at ϕ_0 , with coordinates outside a common probed parameter subset set to zero. For a small step size $\eta > 0$, a first-order expansion gives

$$L_j^{\text{val}}(\phi_0 - \eta \bar{g}_i^{\text{tr}}) \approx L_j^{\text{val}}(\phi_0) - \eta \langle \bar{g}_i^{\text{tr}}, \bar{g}_j^{\text{val}} \rangle. \quad (1)$$

A positive inner product therefore predicts a local reduction in task j ’s loss. To make comparisons tractable, we map the retained gradients to a common d_{CS} -dimensional space using a fixed CountSketch projection (Charikar et al., 2002). The projection is shared across tasks and probe splits; write $\tilde{g} = \text{CS}(\bar{g}) \in \mathbb{R}^{d_{\text{CS}}}$.

For each ordered task pair (i, j) , let D_{ij} denote the gradient alignment and A_{ij} its target-loss-normalized form. These serve as local proxies for absolute and relative loss reduction per unit step size, respectively. Let C_{ij} denote the cosine affinity, which measures agreement between gradient directions (Du et al., 2018). With a small constant $\epsilon > 0$ for numerical stability, we compute these quantities from the compressed gradients:

$$D_{ij} = \langle \tilde{g}_i^{\text{tr}}, \tilde{g}_j^{\text{val}} \rangle, \quad A_{ij} = \frac{D_{ij}}{\max(L_j^{\text{val}}(\phi_0), \epsilon)}, \quad C_{ij} = \frac{D_{ij}}{\max(\|\tilde{g}_i^{\text{tr}}\| \|\tilde{g}_j^{\text{val}}\|, \epsilon)}. \quad (2)$$

We use the cosine affinity matrix $C = [C_{ij}]$ for grouping to reduce the influence of gradient-scale differences across tasks. These affinities are local training proxies; the resulting groups are assessed through teacher and student performance. Probe sizes, parameter selection, and the sketch dimension are specified in Appendix D.

4.2 MIP GROUPING AND SPECIALIST TEACHERS

We use a constrained MIP to turn task affinities into m partially overlapping training groups, balancing within-group compatibility, coverage, and sharing (Wang et al., 2024).

Let S denote the standardized task-affinity matrix, and let μ and σ denote the population mean and standard deviation of all $n(n-1)$ directed off-diagonal entries of C . We compute $S_{ij} = (C_{ij} - \mu)/\sigma$ for $i \neq j$ and set $S_{ii} = 0$. We then define the symmetric task compatibility as $W_{ij} = S_{ij} + S_{ji}$, which combines the two directed affinities.

Let $x_{ig} \in \{0, 1\}$ indicate whether task i belongs to group g , and let k_g denote the number of tasks in that group. Let q_{max} be the maximum number of groups per task and $\lambda \geq 0$ the weight of the

membership penalty. Let $k_{\min}$ and $k_{\max}$ be the lower and upper group-size bounds. We enumerate feasible size profiles $(k_1, \dots, k_m)$ satisfying these bounds. For each profile, we solve

$$\begin{aligned} \max_x \quad & \sum_{g=1}^m \frac{1}{k_g} \sum_{i < j} W_{ij} x_{ig} x_{jg} - \lambda \sum_{i,g} x_{ig} \\ \text{s.t.} \quad & \sum_i x_{ig} = k_g \quad (\forall g), \quad 1 \leq \sum_g x_{ig} \leq q_{\max} \quad (\forall i). \end{aligned} \quad (3)$$

The first term rewards within-group compatibility normalized by group size. The second penalizes total task–group memberships. This penalty is constant within a fixed profile and discourages additional memberships when comparing profiles. The constraints enforce group sizes and task coverage with limited overlap. We linearize the binary products using auxiliary variables and solve each profile with SciPy/HiGHS (Huangfu & Hall, 2018). We retain the returned feasible assignment with the highest objective across profiles.

Each selected group defines the SFT data for specialist T_g . Group membership determines which tasks it learns from, whereas the fixed route determines which tasks it supervises during OPD. Overlap thus permits sharing during teacher training while retaining one specialist route per prompt. The generalist teacher T_A uses the same initialization and SFT recipe on all tasks. Its supervision is used alone in Vanilla OPD and as the anchor in ChemOPD. Appendix D provides the algorithm, configuration, memberships, and routes.

4.3 ROUTED AND ANCHOR-RESIDUAL OPD

Let π_θ denote the student policy, and let π_{ϕ_g} and π_{ϕ_A} denote the frozen policies of specialist T_g and generalist teacher T_A , respectively. We draw a prompt x and its subtask label c from the OPD data distribution $\mathcal{D}_{\text{OPD}}$, generate a student response $y \sim \pi_\theta(\cdot | x)$, and select specialist $g = r(c)$ using the fixed route. Let $\mathcal{M}(x, y)$ denote the valid response positions and h_t the prefix $(x, y_{<t})$ at position t . The routed objective $\mathcal{L}_{\text{rev-KL}}^{(g)}$ measures the expected per-token reverse KL between the student and its specialist (Ma et al., 2026):

$$\mathcal{L}_{\text{rev-KL}}^{(g)}(\theta) = \mathbb{E}_{(x,c), y \sim \pi_\theta} \left[\frac{1}{|\mathcal{M}(x, y)|} \sum_{t \in \mathcal{M}(x, y)} D_{\text{KL}}(\pi_\theta(\cdot | h_t) \| \pi_{\phi_g}(\cdot | h_t)) \right]. \quad (4)$$

For the implemented actor update, we use a top- K teacher-derived advantage (Ma et al., 2026; Gu et al., 2024), where K is the number of student-selected token candidates scored at each prefix. Let $v_{t,k}$ denote the student’s k -th most probable token at prefix h_t , and let $\bar{\pi}_{\theta,t,k}$ denote its probability normalized over these candidates. Let $\hat{A}_{g,t,k}$ denote the specialist-derived advantage that weights the actor update for this candidate. With sg denoting the stop-gradient operator, we compute

$$\bar{\pi}_{\theta,t,k} = \frac{\pi_\theta(v_{t,k} | h_t)}{\sum_{j=1}^K \pi_\theta(v_{t,j} | h_t)}, \quad \hat{A}_{g,t,k} = \text{sg} \left[\bar{\pi}_{\theta,t,k} \log \frac{\pi_{\phi_g}(v_{t,k} | h_t)}{\pi_\theta(v_{t,k} | h_t)} \right]. \quad (5)$$

All model probabilities use temperature-scaled full-vocabulary softmax distributions; only the student weights $\bar{\pi}_{\theta,t,k}$ are normalized over the selected candidates.

To combine generalist and specialist supervision, we score the same prefixes and candidates with both teachers. The generalist’s advantage $\hat{A}_{A,t,k}$ serves as the anchor and is computed by replacing π_{ϕ_g} with π_{ϕ_A} in Equation 5. At OPD update s , let $\alpha_g(s) \in [0, 1]$ denote the specialist weight, shared across response positions and candidates. The combined advantage $\hat{A}_{\text{ChemOPD},t,k}(s)$ adds a weighted specialist correction to the anchor:

$$\hat{A}_{\text{ChemOPD},t,k}(s) = \hat{A}_{A,t,k} + \alpha_g(s)(\hat{A}_{g,t,k} - \hat{A}_{A,t,k}). \quad (6)$$

The specialist residual $\hat{A}_{g,t,k} - \hat{A}_{A,t,k}$ adjusts the generalist’s guidance where the two teachers differ in candidate probabilities. Setting $\alpha_g = 0$ or 1 recovers generalist-only or specialist-only supervision, respectively. We use the combined advantage in the on-policy actor surrogate specified in Appendix I. In the reported configuration, this reduces locally to an advantage-weighted score-function update, with the clipping terms numerically inactive.

We linearly increase the specialist weight to introduce this correction gradually. Let $U \geq 2$ denote the total number of OPD updates and $\alpha_g^{\max}$ the maximum specialist weight. For a warm-up fraction $\rho \in (0, 1]$, let U_{warm} denote the warm-up duration in updates. For $s = 1, \dots, U$, we use

$$U_{\text{warm}} = \max\{2, \lceil \rho U \rceil\}, \quad \alpha_g(s) = \alpha_g^{\max} \min\left\{1, \frac{s-1}{U_{\text{warm}}-1}\right\}. \quad (7)$$

Without warm-up ($\rho = 0$), we set $\alpha_g(s) = \alpha_g^{\max}$ throughout training. The specialist weight controls both the balance between teacher signals and their exposure over training. The reported values and candidate support size are specified in Appendices G and C.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

We evaluate ChemOPD on 20 ChemCoTBench subtasks spanning molecular editing, understanding, optimization, and reaction reasoning (Hao et al., 2026). We use the released structured-CoT training data and task-specific metrics. Only the retrosynthesis evaluation split is changed: malformed references motivate a product-string-disjoint holdout from the training pool. This is not a canonical chemical-identity split; Appendix B gives the data and scoring rules. We train three Qwen3-8B specialists and an additional Qwen3-8B generalist teacher, and distill them into a Qwen3-1.7B student (Yang et al., 2025). Teachers share their initialization and SFT recipe; the student starts from all-task SFT. ChemOPD routes each prompt to one specialist and combines its guidance with the generalist anchor. Appendices C and G provide the implementation details and ablations. We compare MIP specialists with the generalist and with specialists trained on the benchmark’s semantic task families. For student integration, Vanilla OPD uses only the generalist; Benchmark-family OPD uses benchmark-family specialists; and MOPD shares ChemOPD’s specialists, routes, student initialization, prompt stream, and actor-update budget but uses specialists alone (Ma et al., 2026). Random-group OPD keeps the anchor-residual rule and changes the specialist collection. Homogeneous-RR and the DanceOPD-style control vary teacher sampling and batch construction. Full comparison protocols are in Appendix C.1.

5.2 CONSTRUCTING SPECIALIST TEACHERS

Affinity-guided grouping produces task-dependent specialist advantages. The MIP-routed teacher improves 20 of the 31 reported metrics over the generalist teacher, including retrosynthesis FTS from 0.358 to 0.482 (Figure 2). Relative to semantic grouping, optimization scores are unchanged, while several reaction scores improve. Human-defined groupings likewise show gains on some tasks and losses on others (Appendix E). These comparisons support using training relationships to refine specialist task mixtures, rather than expecting specialization to improve every task.

Task organization. The groups contain 8, 8, and 6 tasks (Figure 3). Editing and understanding share a specialist. LogP and solubility appear in both reaction and optimization training groups but are routed only to the optimization specialist during OPD. The reaction teacher therefore learns from tasks beyond those it supervises, illustrating the distinction between training membership and teaching assignments. Coverage and capacity also constrain the grouping; pairwise compatibility is not imposed as a hard requirement. Appendix D provides the full memberships and pairwise affinities. Ten probe draws recover the same grouping up to teacher-label permutation, including both overlap tasks. This supports repeatability of the grouping decision under probe resampling, rather than downstream training variability (Appendix F).

5.3 INTEGRATING SPECIALIST CAPABILITIES

Student performance. ChemOPD improves most reported metrics over Vanilla OPD and Benchmark-family OPD, with gains in editing, optimization, and reaction reasoning (Table 1). With specialists and routes held fixed, it also matches or improves on MOPD in 26 of 31 metrics. Solubility improvement reaches 0.718, compared with 0.490 under generalist-only Vanilla OPD and 0.506 under specialist-only MOPD. The combined training rule can thus outperform either source used alone, although individual tasks still favor single-source supervision. Random-group OPD

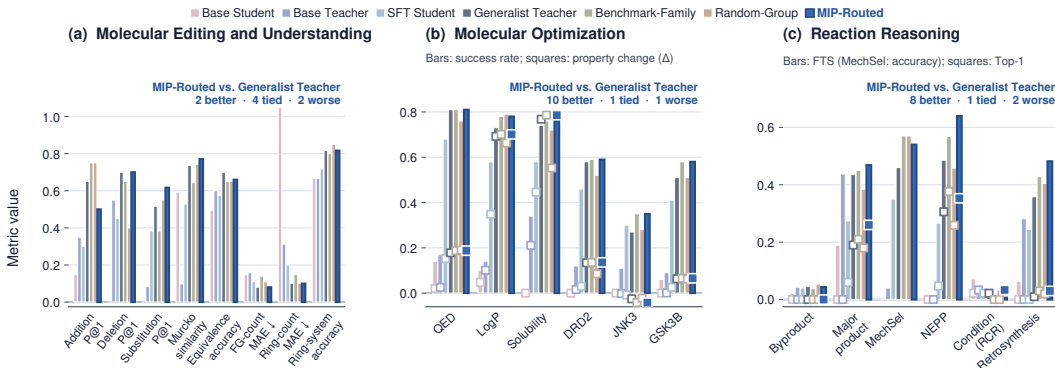


Figure 2: Performance of teacher configurations across benchmark tasks. Base Student and Base Teacher are the pretrained Qwen3-1.7B and Qwen3-8B models, respectively, while SFT Student is the common all-task SFT initializer. Generalist Teacher is trained on all tasks, whereas Benchmark-Family, Random-Group, and MIP-Routed denote three specialist-routing schemes. Bars show the primary metric for each task. Squares show property change (Δ) in panel (b) and Top-1 in panel (c); MechSel is reported by accuracy. Downward arrows mark lower-is-better metrics. Blue annotations summarize the better/tied/worse counts of MIP-Routed relative to Generalist Teacher. Full results are provided in Appendix Table 10.

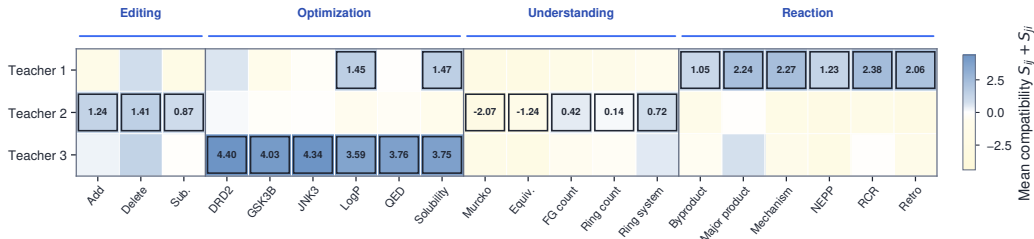


Figure 3: Affinity-based task-group compatibility for the selected MIP grouping. Each cell reports the mean compatibility W_{ij} between the column task and the other tasks assigned to the row teacher. Outlined cells indicate the memberships selected by the MIP. LogP and solubility are shared by Teachers 1 and 3.

retains the same integration rule but performs worse on most metrics, while remaining stronger on LogP. This contrast supports considering teacher construction alongside the integration rule. Homogeneous-RR and DanceOPD-style sampling are competitive on some editing and optimization metrics but generally weaker on reaction reasoning. Changing how teachers are sampled therefore produces different task trade-offs, rather than reproducing the same gains.

Relating student gains to teacher headroom. Positive teacher headroom does not ensure student improvement: on deletion, MOPD regresses from the SFT initializer despite its stronger routed teacher, whereas ChemOPD improves. Headroom recovery summarizes this relationship by dividing student gain by routed-teacher gain over the common initializer, reversing error directions; one denotes recovery of the teacher’s observed gain. To avoid near-zero denominators, we summarize the 26 columns with headroom above 0.02 in native metric units. The reported median rises from 0.37 for MOPD to 0.62 for ChemOPD (Figure 4). The largest recovery increases occur in editing and understanding (0.11 to 0.51) and optimization (0.28 to 0.64). ChemOPD shows its largest recovery gains in families where specialist-only OPD realizes relatively little of the available teacher headroom. Recovery describes student improvement relative to the same specialists, not specialist-specific knowledge transfer; full definitions and family summaries are in Appendix H.1.

5.4 ABLATIONS AND TRAINING BEHAVIOR

Specialist weight and warm-up. We compare ChemOPD with a no-warm-up variant that uses the same final specialist weight of 0.50 from the first update. ChemOPD with warm-up matches or

Table 1: Task-level results across OPD methods. Each panel retains the native metric for each subtask; bold marks the best OPD method in a column before display rounding, and the light-blue row highlights ChemOPD. FG and Ring are lower-is-better; all other metrics are higher-is-better.

(a) Molecule editing & understanding								
Method	EDITING			UNDERSTANDING				
	Add P@1	Delete P@1	Sub. P@1	Murcko Sim.	Equiv. Acc.	FG MAE ↓	Ring MAE ↓	Ring-Sys Sim.
Student, all-task SFT	0.30	0.45	0.38	0.53	0.58	0.11	0.20	0.72
MIP-routed SFT	0.50	0.70	0.62	0.77	0.66	0.08	0.10	0.82
Vanilla OPD	0.35	0.55	0.48	0.55	0.55	0.07	0.15	0.77
Benchmark-family OPD	0.45	0.25	0.40	0.53	0.60	0.10	0.15	0.70
MOPD	0.50	0.20	0.52	0.48	0.61	0.14	0.25	0.77
Homogeneous-RR	0.20	0.55	0.08	0.47	0.67	0.12	0.20	0.78
DanceOPD	0.35	0.65	0.60	0.52	0.64	0.11	0.15	0.73
Random-group OPD	0.30	0.60	0.65	0.53	0.56	0.09	0.15	0.75
ChemOPD	0.35	0.75	0.70	0.58	0.62	0.10	0.15	0.83

(b) Molecular optimization												
Method	QED		LOGP		SOLUBILITY		DRD2		JNK3		GSK3B	
	SR	Δ	SR	Δ	SR	Δ	SR	Δ	SR	Δ	SR	Δ
Student, all-task SFT	0.68	0.153	0.58	0.349	0.58	0.446	0.46	0.030	0.30	-0.008	0.41	0.026
MIP-routed SFT profile	0.81	0.188	0.78	0.701	0.76	0.786	0.59	0.136	0.35	-0.042	0.58	0.066
Vanilla OPD	0.82	0.184	0.74	0.632	0.60	0.490	0.41	0.036	0.27	-0.036	0.41	0.034
Benchmark-family OPD	0.78	0.163	0.58	0.377	0.66	0.632	0.45	0.057	0.26	-0.026	0.43	0.029
MOPD	0.76	0.167	0.70	0.641	0.63	0.506	0.50	0.032	0.18	-0.032	0.42	0.020
Homogeneous-RR	0.74	0.164	0.64	0.557	0.71	0.692	0.57	0.062	0.28	-0.022	0.46	0.023
DanceOPD	0.86	0.194	0.66	0.561	0.70	0.586	0.47	0.053	0.28	-0.030	0.45	0.031
Random-group OPD	0.78	0.165	0.76	0.652	0.69	0.536	0.46	0.040	0.26	-0.034	0.43	0.014
ChemOPD	0.86	0.167	0.68	0.601	0.71	0.718	0.55	0.098	0.29	-0.020	0.46	0.029

(c) Reaction reasoning												
Method	Fwd _{by}		Fwd _{major}		MECHSEL		NEPP		CONDITION		RETRO	
	FTS↑	Top-1	FTS↑	Top-1	Acc.	FTS↑	Top-1	FTS↑	Top-1	FTS↑	Top-1	
Student, all-task SFT	0.040	0.00	0.274	0.06	0.35	0.266	0.05	0.025	0.02	0.245	0.00	
MIP-routed SFT profile	0.044	0.00	0.469	0.26	0.54	0.640	0.35	0.007	0.03	0.482	0.03	
Vanilla OPD	0.043	0.00	0.455	0.09	0.38	0.504	0.07	0.008	0.02	0.318	0.01	
Benchmark-family OPD	0.046	0.00	0.411	0.15	0.42	0.506	0.09	0.000	0.00	0.437	0.02	
MOPD	0.033	0.00	0.398	0.17	0.45	0.462	0.13	0.028	0.03	0.437	0.01	
Homogeneous-RR	0.046	0.00	0.392	0.11	0.00	0.000	0.00	0.000	0.00	0.370	0.00	
DanceOPD	0.023	0.00	0.303	0.09	0.38	0.502	0.08	0.001	0.00	0.386	0.01	
Random-group OPD	0.040	0.00	0.439	0.17	0.37	0.505	0.06	0.000	0.00	0.433	0.00	
ChemOPD	0.051	0.00	0.465	0.18	0.39	0.521	0.13	0.023	0.03	0.443	0.03	

improves 24 of 31 metrics, most clearly in reaction reasoning: retrosynthesis FTS increases from 0.303 to 0.443. QED property gain instead favors constant weighting (0.191 versus 0.167). Thus, matching the final specialist weight is insufficient to reproduce the full integration strategy. The warm-up schedule also controls when and how specialist guidance is introduced during training.

Training behavior. Figure 5 relates the student results to changes in supervision and update scale. Compared with MOPD, ChemOPD has a smaller student-specialist entropy gap, a 29.4% lower extreme-candidate rate, and fewer gradient-norm exceedances after learning-rate warm-up. The entropy channel uses the routed specialist, not a generalist-specialist mixture. These observations

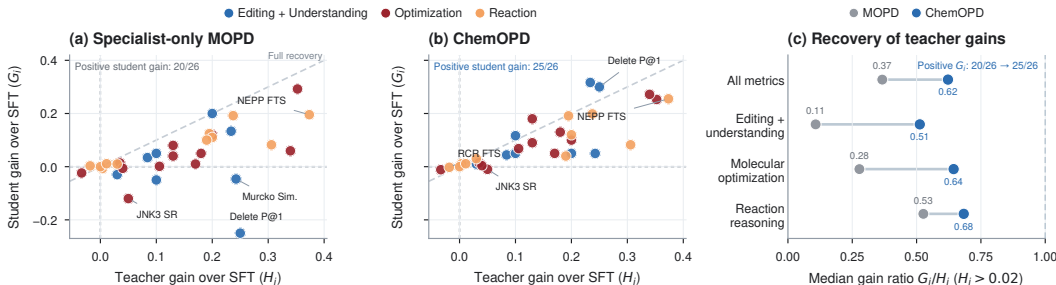


Figure 4: Teacher and student gains over the common SFT initializer. For metric i , H_i is the direction-aligned gain of the routed teacher, and G_i is the corresponding student gain; positive values indicate improvement. Panels (a–b) plot G_i against H_i for specialist-only MOPD and ChemOPD across all 31 metrics. Colors denote task families, and the dashed diagonal $G_i = H_i$ marks full recovery of the teacher gain. Panel (c) reports the median ratio G_i/H_i overall and within each task family for the 26 metrics with $H_i > 0.02$. Gray and blue denote MOPD and ChemOPD, respectively. All gains retain their native metric units; metric columns are descriptive and are not independent trials.

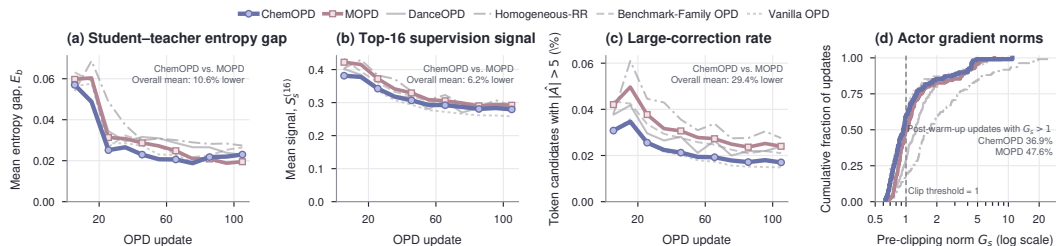


Figure 5: OPD training dynamics. Panels (a–c) show 10-update averages of (a) the student-teacher entropy gap E_b , (b) the signed top-16 supervision signal $S_s^{(16)}$, and (c) the percentage of token candidates with $|\hat{A}| > 5$. Panel (d) shows the empirical CDF of the pre-clipping actor gradient norm G_s ; the dashed line marks $G_s = 1$. Annotations compare ChemOPD with MOPD, which uses the same prompts and specialist routes. See Appendix I for definitions.

do not by themselves identify the causal role of each training signal. However, they show that ChemOPD introduces specialist supervision with a more moderate training signal while maintaining broader capability integration. Diagnostic definitions are provided in Appendix I. Checkpoint trajectories provide a complementary view (Appendix H.2). MOPD improves the reaction summaries faster early in training, whereas ChemOPD maintains a higher understanding summary and finishes ahead on both reaction summaries. Thus, faster early gains in one task family do not necessarily yield a stronger final cross-task profile.

6 CONCLUSION

We introduced ChemOPD, a framework for organizing teacher specialization and integrating chemical capabilities into a unified student. SFT-gradient affinities and constrained mixed-integer programming determine which tasks specialists learn together. Anchor-residual OPD then combines their guidance with generalist supervision, gradually increasing the specialist contribution. Experiments on ChemCoTBench show task-dependent teacher gains from affinity-guided grouping and student improvements over generalist-only and specialist-only distillation across multiple chemical tasks. Component ablations favor the full training rule over specialist-only supervision and constant mixing on most reported metrics, although individual tasks can favor the alternatives. Together, these findings support considering not only how teachers develop specialized strengths, but also how those strengths complement generalist supervision when training a unified chemical reasoner.

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A AI USE STATEMENT

In preparing this manuscript, generative AI tools were used for language editing, manuscript restructuring, and consistency checking. All scientific claims, equations, experimental results, citations, and AI-assisted revisions were reviewed by the authors. The authors take responsibility for the final content of the submission.

B BENCHMARK TASKS, METRICS, AND DATA PROTOCOL

Table 2: 20 chemical reasoning subtasks used in the study. Detailed task semantics and metric calculations are given in Appendix B.2.

Family	Subtasks	Primary evaluation dimensions
Editing	Add, Delete, Substitution	P@1 for a valid output satisfying the requested functional-group change
Understanding	Murcko scaffold, Equivalence, functional-group count, ring count, ring-system recognition	Murcko Morgan-Tanimoto; equivalence/ring-system accuracy; FG/ring MAE
Optimization	QED, LogP, solubility, DRD2, JNK3, GSK3B	Property gain Δ and success rate $SR(\Delta > 0)$; validity/scaffold diagnostics
Reaction	Fwd _{by} , Fwd _{major} , MechSel, NEPP, Condition/RCR, Retro	Top-1 chemical match and FTS for SMILES tasks; MechSel accuracy

B.1 DATASET CONSTRUCTION AND REACTION PROTOCOL

We use the released ChemCoTBench training and evaluation protocol for all subtasks except retrosynthesis, for which we construct a product-string-disjoint holdout from the released training pool as described below.

B.2 TASK SEMANTICS AND METRIC CALCULATIONS

We follow the task taxonomy and structured-output evaluation of ChemCoTBench, using the 20 tasks in our protocol. Editing asks the model to return a molecule after adding, deleting, or substituting a specified functional group. Understanding covers Murcko-scaffold extraction, SMILES equivalence (whether two strings denote the same molecule), functional-group and ring counting, and binary ring-system recognition. Optimization asks for a molecule that improves one requested property: QED (drug-likeness) (Bickerton et al., 2012), LogP (lipophilicity), aqueous solubility (ESOL) (Delaney, 2004), or a target-specific oracle score (DRD2, JNK3, or GSK3B). Reaction reasoning contains forward byproduct and major-product prediction, mechanism-route selection (MechSel), next-elementary-step product prediction (NEPP), reaction condition recommendation (RCR/Condition), and single-step retrosynthesis (Retro). The forward views predict products from reactants, NEPP predicts the next-step product, MechSel selects a reaction route, RCR predicts reaction conditions, and Retro infers reactants from a product. The split and view counts used here are defined below.

For a task with N evaluation examples, let V denote the examples with a valid or recognized answer. Whenever validity is reported, it is

$$\text{Validity} = \frac{|V|}{N}. \quad (8)$$

For categorical subtasks, we report accuracy

$$\text{Acc} = \frac{1}{|S|} \sum_{i \in S} \mathbf{1}[\hat{y}_i = y_i], \quad (9)$$

after the task-specific normalization used by the evaluator. The scored set S is the recognized-answer subset for SMILES equivalence (a valid Yes/No answer), but is the full evaluation set (with an unparseable choice counted as incorrect) for ring-system recognition and multiple-choice MechSel. Editing P@1 uses the total evaluation set as its denominator, with a success requiring a parseable output whose functional-group count changes satisfy the requested Add, Delete, or Substitution operation; equivalently, $\text{P@1} = N_{\text{success}}/N$, where N_{success} is the number of outputs passing that operation check. For functional-group and ring counts, we instead report mean absolute error over parseable outputs,

$$\text{MAE} = \frac{1}{|V|} \sum_{i \in V} |\hat{c}_i - c_i|, \quad (10)$$

and report the valid-output rate separately; thus editing P@1 is a one-output operation-success rate rather than an exact-SMILES rate. Murcko extraction (Bemis & Murcko, 1996) is scored by the Tanimoto similarity of radius-2 Morgan fingerprints (1,024 bits) (Rogers & Hahn, 2010) for the predicted and reference scaffolds; an identical scaffold therefore scores one. For binary fingerprints a, b , the Tanimoto score is $T(a, b) = |a \cap b| / |a \cup b|$, where the sets denote active bits.

For molecule optimization, we retain the released ChemCoTBench property oracles and aggregation rules, while a local ID-aligned wrapper extracts the generated molecule from each model response. Let N be the total number of evaluation examples for one property, p the requested property, and m_i and $\hat{m}_i$ the source and generated molecules. For a successfully scored example, define the raw improvement

$$\delta_i = p(\hat{m}_i) - p(m_i).$$

If a prediction is missing, cannot be extracted, is invalid, or cannot be scored by the property oracle, its improvement is set to zero. Thus the zero-padded vector $\mathbf{x} = (x_1, \dots, x_N)$ contains the raw δ_i values for scorable examples and zeros otherwise. Let $L = P_5(\mathbf{x})$ and $U = P_{95}(\mathbf{x})$ be the 5th and 95th percentiles and define $\tilde{x}_i = \text{clip}(x_i, L, U)$. The two reported optimization quantities are

$$\begin{aligned} \bar{\Delta}_{\text{display}} &= \frac{1}{N} \sum_{i=1}^N \tilde{x}_i, \\ \text{SR} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}[x_i > 0]. \end{aligned} \quad (11)$$

Accordingly, the displayed mean property change uses the winsorized, zero-padded improvements, whereas success rate uses the raw zero-padded improvements with the strict criterion $x_i > 0$. Both use all evaluation examples for that property as the denominator. These rules follow the released benchmark evaluator; the local wrapper adds robust output extraction, ID alignment, and per-sample diagnostics without changing the released property oracles or aggregate definitions. When reported, hard scaffold consistency is exact equality of the source and generated Murcko scaffolds, and soft consistency is their scaffold fingerprint Tanimoto. For reaction SMILES generation, Top-1 is an RDKit-parseable chemical match (InChI equality in the evaluator), while the fingerprint similarity score (FTS) is the validity-gated mean of three Tanimoto similarities:

$$\text{FTS} = \text{Validity} \frac{T_{\text{RDK}} + T_{\text{MACCS}} + T_{\text{Morgan}}}{3}. \quad (12)$$

The three fingerprints compare the generated and reference molecules; the released evaluator averages their per-example similarities (invalid parses contribute zero) and gates the result by the aggregate validity rate, where validity is the fraction of RDKit-parseable predictions. Higher values are better for all primary metrics except FG/Ring MAE, for which lower values are better. Ring-Sys Sim denotes the binary ring-system accuracy described above. These definitions follow the released benchmark evaluator; our protocol-specific counts, filtering, and retrosynthesis holdout are given in the remainder of this section.

Overall corpus. Each SFT instance pairs a benchmark prompt with its released structured-CoT assistant target; prompt tokens provide context but are masked from the SFT loss. Table 3 gives the resulting task-instance counts. The evaluation protocol contains 1,595 nominal task views originating from 1,495 source records. The only source-to-multiple-view expansion is forward synthesis: each of the 100 released forward records is exposed once as a major-product view and once as a byproduct view. The paired views share the same prompt and source identifier but have distinct scalar targets and view-suffixed evaluation identifiers. Nineteen byproduct views have empty released references and are omitted by the reaction evaluator, leaving 1,576 reference-bearing per-sample evaluation rows. This number is not a universal denominator for every reported metric, because task-specific evaluators may condition aggregation on successful output parsing or other benchmark-defined validity rules. For every non-retrosynthesis task, we preserve the released evaluation data; the only split change is the deterministic retrosynthesis holdout described below.

Table 3: SFT training instances and evaluation task views by benchmark family.

Family	SFT training	Evaluation
Molecular editing	4,497	100
Molecular understanding	7,718	320
Molecular optimization	5,587	600
Reaction reasoning	10,371	575
Total	28,173	1,595

The prompt-only OPD file contains one row per SFT instance. Under the current 1,024-token prompt limit, runtime length filtering retains 28,089 prompt rows. With a prompt batch size of 256 and incomplete batches dropped, one OPD epoch contains 109 optimizer batches and samples 27,904 prompts.

Normalization of released reaction tasks. For non-retrosynthesis reaction tasks, we preserve the released training and evaluation data. The forward task is represented differently in SFT and evaluation. For SFT, we retain the separately released structured-CoT corpora as provided, comprising 2,788 major-product training rows and 890 byproduct training rows. These corpora are not formed by duplicating a common set of forward reactions: their identifiers and reaction coverage differ, and their prompts explicitly request major-product or byproduct prediction, respectively. Accordingly, the reported SFT counts are training rows rather than counts of distinct reactions. For evaluation, by contrast, each of the 100 released forward records contains both targets and is exposed as two scalar-target views sharing the same prompt, one for the major product and one for the byproduct. Table 4 reports the resulting counts.

Retrosynthesis test set re-split. Inspection of the released retrosynthesis test snapshot shows that 81 of the 100 reference targets correspond to multiple reactants whose SMILES strings are concate-

Table 4: Reaction-task instance counts under the current protocol. Each released forward-synthesis example contributes one major-product and one byproduct view.

Subtask	Train	Eval.	Scored
Byproduct prediction	890	100	81
Major-product prediction	2,788	100	100
Mechanism selection	229	100	100
NEPP	1,257	85	85
RCR	3,142	90	90
Retrosynthesis	2,065	100	100
Total	10,371	575	556

nated without the . component separators required by the task output format; the other 19 contain a single reactant. For the multi-reactant targets, the molecular boundaries cannot be recovered unambiguously from the concatenated strings alone. Heuristically inserting separators would therefore modify the ground-truth annotations rather than reliably restore them. We construct a new evaluation holdout from the 2,165 records in the released retrosynthesis structured-CoT training pool, whose metadata preserves the reactant targets. We select 100 records for evaluation and retain the remaining 2,065 records for training.

Construction of the product-string-disjoint holdout. For each of the 2,165 source training records, we read raw data, collect its dot-separated fragments, strip leading and trailing whitespace, discard empty fragments, sort the remaining fragment strings lexicographically, and rejoin them with . to obtain a normalized product-string key. Records sharing the same key are assigned to the same partition. Evaluation examples are selected only from normalized product-string keys that occur once in the source training pool. Consequently, once a record is selected for evaluation, its normalized product-string key does not appear in the retained training set, thereby preventing train-test overlap at the normalized product-string level. Under the 100-record evaluation budget, the deterministic selection first maximizes reaction-class coverage among the eligible singleton groups. This covers 82 of the 87 source reaction classes, corresponding to all classes with at least one singleton candidate. The remaining 18 positions are filled according to class-frequency deficits, with at most one additional record per class during this stage. This construction guarantees product-string disjointness only under the normalization described above, which accounts for fragment ordering and surrounding whitespace.

Table 5: Construction of the retrosynthesis holdout.

Statistic	Count
Source records	2,165
Normalized product-string groups	1,818
Source reaction classes	87
Singleton product-string groups	1,676
Classes with singleton candidates	82
Training records	2,065
Evaluation records	100
Evaluation product groups / exact queries	100 / 100
Reaction classes represented in evaluation	82

We verify that none of the 100 held-out examples overlap with any training or affinity-probe data used in our pipeline by record ID, exact retrosynthesis prompt, or normalized product-string key, ruling out train-test leakage under these matching criteria.

C IMPLEMENTATION AND COMPARISON PROTOCOLS

Tables 6 and 7 summarize the SFT and OPD configurations used for the completed experiments; affinity-probe and MIP settings are reported with the grouping in Table 9. We use the same data-shuffling protocol across methods, and evaluate each model with temperature-zero decoding. This

setup controls the main sources of run-to-run variation and keeps the comparison conditions consistent across methods. The tables retain the principal settings shared across methods. The following subsections specify the comparisons and method-specific batch construction.

C.1 COMPARISON PROTOCOLS

Teacher construction. All-task training fits one teacher to the full task collection, whereas benchmark-family grouping trains specialists on the benchmark’s semantic families. The MIP construction uses affinity-guided groups with limited overlap. Human-defined controls combine the understanding and optimization families, understanding and reaction, or editing and optimization. The capacity-matched random construction preserves the group capacity but assigns tasks randomly. Human controls are compared only on tasks they cover; missing entries are not failures. Complete scores and the human-control visualization appear in Appendix E.

Student integration. Vanilla OPD uses the all-task teacher, and benchmark-family OPD uses its family specialists. Specialist-only MOPD (Ma et al., 2026) shares ChemOPD’s SFT student initialization, MIP specialist checkpoints, fixed routes, prompt stream, and actor-update budget, but omits the anchor. Random-group OPD retains the ChemOPD objective with a randomly constructed specialist collection. We use it to compare teacher collections under the same integration objective.

Homogeneous-RR and the DanceOPD-style control change how routed training units are assembled into an update. The latter adapts a sampling strategy from Zhou et al. (2026), not its flow-matching image-model objective. The exact route sampling and task exposures are specified below. Equal update, prompt, and rollout budgets do not make these controls task-exposure matched to ChemOPD and MOPD.

C.2 BATCH CONSTRUCTION AND TEACHER SCORING

Method-specific OPD batch construction. For ChemOPD and MOPD, runtime prompt-length filtering retains 28,089 of the 28,173 prepared prompt rows. With batch size 256 and drop-last batching, the standard shuffled stream therefore yields 109 distributed optimizer updates and 27,904 prompt draws. Four rollouts per prompt produce 111,616 rollout sequences in total. The resulting task-family exposure is 10,222 reaction prompts, 12,138 editing/understanding prompts, and 5,544 optimization prompts. Homogeneous-RR preserves the same total update, prompt, and rollout budgets, but each complete 256-prompt update comes from a single specialist pool. The three specialists are visited in strict round-robin order, yielding 37/36/36 updates and 9,472/9,216/9,216 prompts for reaction, editing/understanding, and optimization, respectively.

The DanceOPD baseline also preserves 109 updates and 256 prompts per update, but partitions each update into four contiguous 64-prompt homogeneous route slots. Each slot independently samples one route, with replacement, from the all-task teacher and the three specialists. The four slots are concatenated, scored by their selected teachers, restored to batch order, and jointly passed to one actor optimizer update; they are not four separate backward/optimizer steps. Thus, both sampling-based controls match ChemOPD/MOPD in the total numbers of updates, prompts, and rollouts, but distribute the training prompts differently across task families.

The DanceOPD comparison retains only the aspects that transfer naturally to this language-model setting-on-policy rollouts, hard teacher routing, route sampling with replacement, and aggregation of several routed units into one update. It does not reproduce the original flow-matching velocity field, diffusion trajectory-state queries, timestep sampling, velocity-MSE objective, LoRA teacher-field composition, or off-policy teacher-trajectory branch.

Table 6: Core supervised fine-tuning settings. Microbatch size, gradient accumulation, and gradient checkpointing were adjusted to model size and GPU allocation; the optimizer and data-processing settings below were shared.

Hyperparameter	Value
Teacher initialization	Qwen3-8B-Base
Student initialization	Qwen3-1.7B-Base
Fine-tuning scope	Full-parameter
Training objective	Assistant-token causal LM loss; prompt masked
Epochs	3
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 0.01
Learning rate	2×10^{-5} , constant
Maximum sequence length	4096 tokens
Numerical precision	bfloat16
Gradient clipping	Global norm 1.0
Distributed backend	DDP
Random seed	42

Table 7: OPD hyperparameters shared by the completed methods. The method-specific batch construction used by Homogeneous-RR and DanceOPD is described in Appendix C.2; the implemented distillation signal and actor objective are defined in Appendix I.

Hyperparameter	Value
Student initialization	All-task SFT Qwen3-1.7B
Training budget	109 distributed optimizer updates
Prompt batch size	256
Rollouts per prompt	4
Actor minibatch	256 prompts / 1,024 rollout sequences globally
Computational microbatching	Dynamic per-GPU packing; 4,096-token budget
Prompt/response limit	1024 / 1024 tokens
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 0.01
Peak learning rate	1.2×10^{-5}
Learning-rate schedule	Cosine; 5% warm-up; minimum ratio 0.1
Gradient clipping	Global norm 1.0
Distillation signal	Detached student-top-16 reverse-KL-style discrepancy
Student support weighting	Top-16 by student; q^S renormalized on support
Teacher log-probabilities	Full-softmax values gathered on student top-16
Actor objective	On-policy clipped policy-gradient surrogate (PPO-style)
Loss aggregation	Candidate sum; masked response-token mean within dynamic microbatches
Rollout/teacher temperature	0.7/0.7
Rollout truncation	$p = 1.0$; no top- k truncation
External correctness reward	None
Thinking mode	Disabled
Data order and seed	Shuffled; seed 42
Reported checkpoint	Step 109

D TASK GROUPING AND CONSTRUCTION SETTINGS

Algorithm 1 specifies the profile-enumeration and MILP procedure. The experiments use $m = 3$, group sizes from four to eight, $q_{\max} = 3$, and membership penalty $\lambda = 0.1$. The selected 8/8/6 profile and fixed routes are listed in Table 8. The table provides the complete task assignment and fixed OPD routes; LogP and solubility are the only shared memberships. Table 9 collects the probe and solver settings used to obtain this assignment.

Pairwise compatibility within the selected groups. Figure 6 expands the task-group averages in Figure 3 into the pairwise compatibility scores W_{ij} within each selected teacher group. This view

Algorithm 1 Soft-overlap MIP task grouping**Require:** Cosine affinity $C \in \mathbb{R}^{n \times n}$, group count m , size bounds, membership bounds, overlap penalty λ **Ensure:** Groups $G_1, \dots, G_m$

```

1:  $S \leftarrow \text{ZSCOREOFFDIAGONAL}(C)$ 
2:  $\mathcal{K} \leftarrow \text{FEASIBLEPROFILES}(n, m, k_{\min}, k_{\max}, q_{\max})$ 
3:  $(z^*, x^*) \leftarrow (-\infty, \emptyset)$ 
4: for  $(k_1, \dots, k_m) \in \mathcal{K}$  do
5:   Create binary memberships  $x_{ig}$  and pair variables  $y_{ijg}$ 
6:   Add exact group-size and task-coverage constraints
7:   Add  $y_{ijg} = x_{ig}x_{jg}$  through standard linear constraints
8:    $z \leftarrow \sum_g k_g^{-1} \sum_{i < j} W_{ij} y_{ijg} - \lambda \sum_{i,g} x_{ig}$ 
9:    $(\hat{z}, \hat{x}) \leftarrow \text{SOLVEMILP}(\max z)$ 
10:  if  $\hat{x}$  is feasible and  $\hat{z} > z^*$  then
11:     $(z^*, x^*) \leftarrow (\hat{z}, \hat{x})$ 
12:  end if
13: end for
14: return  $G_g = \{i : x_{ig}^* = 1\}$  for all  $g$ 

```

Table 8: Affinity-guided soft-overlap groups used by the current reaction protocol experiments. LogP and Solubility occur in both Teacher 1 and Teacher 3; their fixed OPD route is Teacher 3.

Teacher	SFT subtasks	Rows	Fixed OPD route
Teacher 1	Opt/LogP, Opt/Solubility, Reaction/Byproduct, Reaction/Major product, Reaction/Mechanism selection, Reaction/NEPP, Reaction/RCR, Reaction/Retrosynthesis	13,177	All Reaction tasks
Teacher 2	Edit/Add, Edit/Delete, Edit/Substitution, Understanding/Murcko scaffold, Understanding/Equivalence, Understanding/Functional-group count, Understanding/Ring count, Understanding/Ring-system scaffold	12,215	Editing and Understanding
Teacher 3	Opt/DRD2, Opt/GSK3B, Opt/JNK3, Opt/LogP, Opt/QED, Opt/Solubility	5,587	All Optimization tasks

Table 9: Affinity-probe and MIP settings used to construct the three ChemOPD specialists. The selected profile is a practical solver output rather than a claim of a unique globally optimal grouping.

Hyperparameter	Value
Probe backbone	Qwen3-8B-Base
Probe examples per task	100 train / 50 validation
Probe batch/sequence length	1 / 4096 tokens
Gradient representation	Last 2 transformer layers; LM head excluded
CountSketch dimension	8192
Probe precision and seed	bfloat16; 42
Affinity and preprocessing	Cosine; off-diagonal z -score
Number of task groups	3
Group-size bounds	4–8 tasks
Membership bounds	1–3 groups per task
Extra-membership bounds	Minimum 0; maximum set by capacity
Overlap penalty λ	0.1 per task–group membership
Objective normalization	Group size, $1/ G_g $
Symmetric compatibility	$W_{ij} = S_{ij} + S_{ji}$
Solver	SciPy/HiGHS MILP
Time/gap limits	300 s / relative gap 0.01
Selected group sizes	8/8/6 (two overlapping tasks)
Returned solve statistics	222.9 s; relative gap 0.00173

shows the variation among task pairs within the same group. The complete membership constraints and solver settings are reported above.

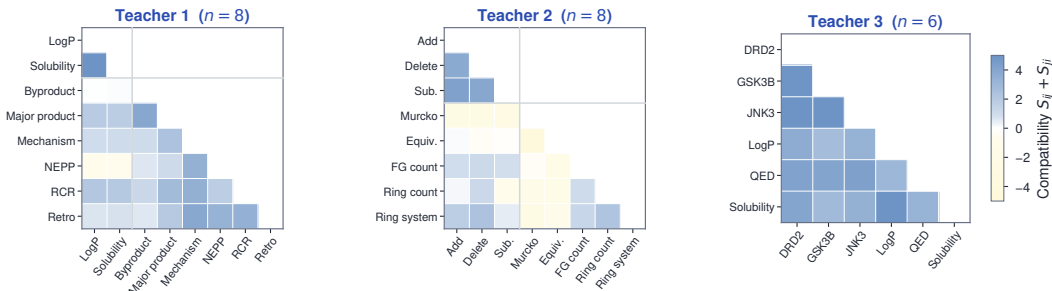


Figure 6: Pairwise compatibility within the three MIP-selected teacher groups. Each panel shows $W_{ij} = S_{ij} + S_{ji}$ for every unique off-diagonal task pair within one group. Positive and negative values indicate above- and below-average compatibility, respectively.

E FULL TEACHER-CONSTRUCTION RESULTS

Table 10 reports the complete task-level results for the teacher constructions compared in Section 5.2. For the MIP and benchmark-family groupings, each task is evaluated using its predefined teacher route. Human-defined controls are evaluated only on the tasks covered by their respective groups, and blank entries indicate uncovered tasks rather than evaluation failures.

Human-defined grouping controls. The Understanding+Optimization, Understanding+Reaction, and Editing+Optimization controls test whether intuitive family combinations produce comparable specialists. They often strengthen one capability while weakening another. For example, Understanding+Reaction improves several reaction metrics relative to the all-task teacher but gives up understanding performance. MIP matches or improves on most covered human-control metrics, although the advantages are not uniform (Figure 7). The curves preserve native metric scales and leave uncovered axes open.

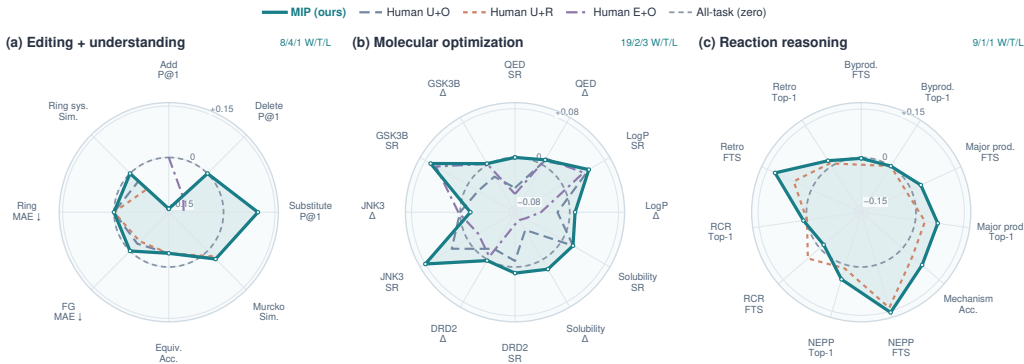


Figure 7: Direction-aligned teacher gains from Table 10. The dashed ring is the generalist teacher; outward is better, with MAE signs reversed. Human-control curves remain open for uncovered tasks, while MIP-routed follows its fixed route. Teal labels report pooled pairwise comparisons between MIP-routed and all available human-control results. Each metric contributes one comparison for every human control that covers it. Metrics retain native units and panel-specific scales.

Table 10: Full task-level results for the teacher constructions. The MIP and benchmark-family profiles use predefined task-to-teacher routes. The shaded row shows the routed profile of the MIP specialists used by ChemOPD. Blank entries indicate tasks not covered by the corresponding human-defined controls.

(a) Molecule editing & understanding										
Configuration	Model	EDITING			UNDERSTANDING					
		Add P@1	Delete P@1	Sub. P@1	Murcko Sim.	Equiv. Acc.	FG MAE ↓	Ring MAE ↓	Ring-Sys Sim.	
Student, raw	Qwen3-1.7B	0.15	0.00	0.00	0.59	0.49	0.15	1.05	0.67	
Teacher, raw	Qwen3-8B	0.35	0.55	0.08	0.10	0.60	0.16	0.31	0.67	
Student, all-task SFT	Qwen3-1.7B	0.30	0.45	0.38	0.53	0.58	0.11	0.20	0.72	
Teacher, all-task SFT	Qwen3-8B	0.65	0.70	0.52	0.74	0.70	0.08	0.10	0.82	
Benchmark-family SFT	Qwen3-8B	0.75	0.65	0.38	0.64	0.65	0.14	0.15	0.80	
Random-group SFT	Qwen3-8B	0.75	0.40	0.55	0.74	0.65	0.11	0.10	0.85	
MIP-routed SFT	Qwen3-8B	0.50	0.70	0.62	0.77	0.66	0.08	0.10	0.82	
Human U+O SFT	Qwen3-8B	–	–	–	0.77	0.66	0.11	0.10	0.78	
Human U+R SFT	Qwen3-8B	–	–	–	0.76	0.66	0.12	0.10	0.75	
Human E+O SFT	Qwen3-8B	0.65	0.60	0.40	–	–	–	–	–	

(b) Molecular optimization													
Configuration	Model	QED		LOGP		SOLUBILITY		DRD2		JNK3		GSK3B	
		SR	Δ	SR	Δ	SR	Δ	SR	Δ	SR	Δ	SR	Δ
Student, raw	Qwen3-1.7B	0.14	0.020	0.10	0.048	0.02	0.000	0.00	0.000	0.01	0.000	0.06	0.001
Teacher, raw	Qwen3-8B	0.17	0.025	0.14	0.101	0.34	0.211	0.12	0.016	0.11	-0.002	0.09	0.001
Student, all-task SFT	Qwen3-1.7B	0.68	0.153	0.58	0.349	0.58	0.446	0.46	0.030	0.30	-0.008	0.41	0.026
Teacher, all-task SFT	Qwen3-8B	0.81	0.178	0.73	0.692	0.74	0.768	0.58	0.134	0.27	-0.025	0.51	0.063
Benchmark-family SFT	Qwen3-8B	0.81	0.188	0.78	0.701	0.76	0.786	0.59	0.136	0.35	-0.042	0.58	0.066
Random-group SFT	Qwen3-8B	0.76	0.190	0.79	0.663	0.72	0.553	0.52	0.085	0.28	-0.023	0.51	0.062
MIP-routed SFT	Qwen3-8B	0.81	0.188	0.78	0.701	0.76	0.786	0.59	0.136	0.35	-0.042	0.58	0.066
Human U+O SFT	Qwen3-8B	0.76	0.185	0.78	0.672	0.76	0.712	0.57	0.114	0.30	-0.025	0.49	0.041
Human E+O SFT	Qwen3-8B	0.75	0.190	0.77	0.645	0.67	0.693	0.51	0.128	0.25	-0.022	0.57	0.065

(c) Reaction reasoning													
Configuration	Model	Fwd _{by}		Fwd _{major}		MECHSEL	NEPP		CONDITION		RETRO		
		FTS↑	Top-1	FTS↑	Top-1		Acc.	FTS↑	Top-1	FTS↑	Top-1	FTS↑	
Student, raw	Qwen3-1.7B	0.021	0.00	0.189	0.00	0.00	0.000	0.00	0.072	0.02	0.064	0.00	
Teacher, raw	Qwen3-8B	0.043	0.00	0.437	0.00	0.04	0.000	0.00	0.042	0.03	0.282	0.00	
Student, all-task SFT	Qwen3-1.7B	0.040	0.00	0.274	0.06	0.35	0.266	0.05	0.025	0.02	0.245	0.00	
Teacher, all-task SFT	Qwen3-8B	0.046	0.00	0.435	0.19	0.46	0.486	0.31	0.022	0.02	0.358	0.01	
Benchmark-family SFT	Qwen3-8B	0.037	0.00	0.451	0.21	0.57	0.568	0.38	0.005	0.00	0.428	0.03	
Random-group SFT	Qwen3-8B	0.054	0.00	0.385	0.18	0.57	0.458	0.26	0.034	0.00	0.405	0.02	
MIP-routed SFT	Qwen3-8B	0.044	0.00	0.469	0.26	0.54	0.640	0.35	0.007	0.03	0.482	0.03	
Human U+R SFT	Qwen3-8B	0.024	0.00	0.419	0.22	0.50	0.625	0.32	0.071	0.02	0.419	0.02	

F ROBUSTNESS TO AFFINITY-PROBE RESAMPLING

Section 5.2 evaluates whether affinity-guided grouping produces useful specialist teachers. Here, we study a complementary question: Does the estimated affinity structure remain stable when the probe examples change? We also test whether these changes lead the MIP to select different task groups. This analysis checks whether the grouping reported in the main text depends on one particular probe sample.

Resampling protocol. We use the affinity matrix and its 8/8/6 grouping as the reference. We denote this reference run by $b = 0$. We then create $B = 10$ new probe draws, indexed by $b \in \{1, \dots, B\}$, from our training split. For each task in each draw, we sample 100 training-probe examples and 50 validation-probe examples without replacement. Within a draw, the two probe subsets are disjoint by normalized prompt. For retrosynthesis, they are also disjoint by normalized product-string key. Examples may be reused across different draws.

We keep the Qwen3-8B-Base checkpoint, probed gradient layers, and CountSketch map fixed. We also keep the MIP objective and constraints fixed. All ten audit runs use a 600-second solver limit and a relative gap of 0.001. This setting is tighter than the solver setting used for the original

grouping and reduces solver noise in the audit. Across the ten audit runs, only the probe examples change. We do not retrain any specialist teacher or student model.

Affinity-stability metrics. For run b , let $C^{(b)}$ be the directed cosine-affinity matrix defined in Equation 2. Let $S^{(b)}$ be the off-diagonal z -score of $C^{(b)}$, computed in the same way as in the main grouping method. We define the symmetric task-compatibility matrix as

$$W^{(b)} = S^{(b)} + S^{(b)\top}. \quad (13)$$

Let $w^{(b)} = \text{vec}_{\Delta}(W^{(b)})$ contain the upper-triangular entries of $W^{(b)}$. We have 20 tasks, so $w^{(b)}$ contains $\binom{20}{2} = 190$ task-pair values. We compare each resampled vector with the reference vector using

$$\begin{aligned} \rho_{\text{rank}}^{(b)} &= \text{Spearman}(w^{(0)}, w^{(b)}), \\ \rho_{\text{value}}^{(b)} &= \text{Pearson}(w^{(0)}, w^{(b)}). \end{aligned} \quad (14)$$

The Spearman correlation measures whether the task-pair ordering is preserved. The Pearson correlation measures whether the compatibility values are preserved.

We next measure local stability. For task i , let $N_3^{(b)}(i)$ be the set of its three highest-scoring neighbors under $W^{(b)}$. Let $J_i^{(b)}$ compare this set with the reference neighbors. We average this score over the $n = 20$ tasks and denote the result by $\bar{J}_3^{(b)}$. We also examine task pairs with large reference compatibility. Let $\mathcal{E}_s$ contain the 48 pairs in the top quartile of $|W_{ij}^{(0)}|$. We denote their sign agreement by $A_{\text{sign}}^{(b)}$. These metrics are

$$\begin{aligned} J_i^{(b)} &= \frac{|N_3^{(0)}(i) \cap N_3^{(b)}(i)|}{|N_3^{(0)}(i) \cup N_3^{(b)}(i)|}, \\ \bar{J}_3^{(b)} &= \frac{1}{n} \sum_{i=1}^n J_i^{(b)}, \\ A_{\text{sign}}^{(b)} &= \frac{1}{|\mathcal{E}_s|} \sum_{(i,j) \in \mathcal{E}_s} \mathbf{1}[\text{sgn } W_{ij}^{(0)} = \text{sgn } W_{ij}^{(b)}]. \end{aligned} \quad (15)$$

The sign in $A_{\text{sign}}^{(b)}$ is the sign of the standardized compatibility $W_{ij}^{(b)}$. It is not the sign of the raw gradient dot product.

Grouping-stability metrics. Let $x^{(0)}$ be the reference MIP assignment and let $x^{(b)}$ be the assignment obtained from draw b . As in Equation 3, $x_{ig} = 1$ means that task i belongs to group g . Teacher labels can be permuted without changing the grouping. We therefore compare task co-membership instead of teacher labels. Let $E(x)$ be the set of task pairs that share at least one group under assignment x . We measure grouping agreement using

$$J_{\text{group}}^{(b)} = \frac{|E(x^{(0)}) \cap E(x^{(b)})|}{|E(x^{(0)}) \cup E(x^{(b)})|}. \quad (16)$$

Let $F_b(x)$ be the MIP objective in Equation 3, evaluated using $S^{(b)}$. We define the objective gap between the re-solved assignment and the reference assignment as

$$\Delta F_b = F_b(x^{(b)}) - F_b(x^{(0)}). \quad (17)$$

A zero gap means that re-solving the MIP does not improve on the reference assignment for that probe draw. Finally, we compare the reference grouping with matched random assignments. For each draw, we randomly permute the task labels in $x^{(0)}$ 10,000 times. This preserves the 8/8/6 group sizes and the overlap structure. Let μ_b^{rand} and σ_b^{rand} be the mean and standard deviation of the resulting objective values. We report

$$Z_b = \frac{F_b(x^{(0)}) - \mu_b^{\text{rand}}}{\sigma_b^{\text{rand}}}. \quad (18)$$

Thus, Z_b measures how far the reference grouping lies above the matched random baseline.

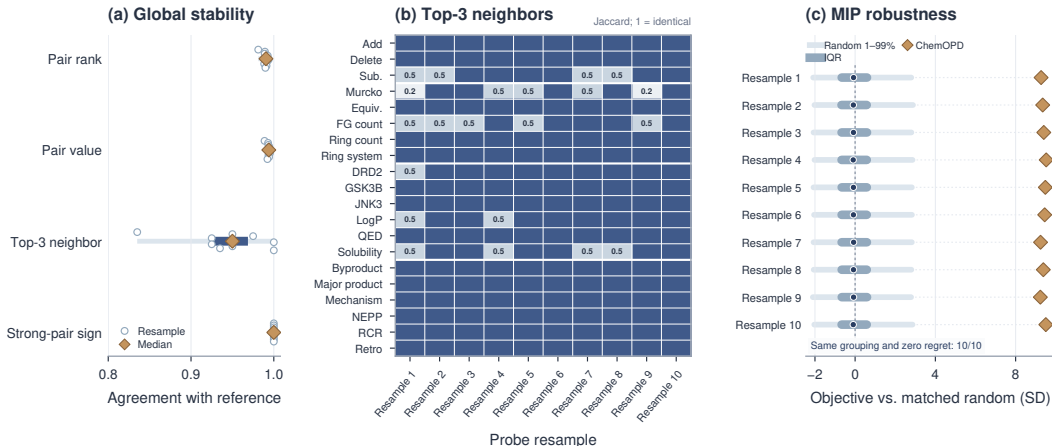


Figure 8: Stability of affinity estimation and MIP grouping under probe resampling. (a) Agreement between each resampled affinity matrix and the reference. Open circles show the ten draws. Light bars show the full range, dark bars show the interquartile range, and diamonds show the median. The top-3 neighbor score is averaged over tasks. (b) Task-level top-3 neighbor Jaccard scores. A value of one means that all three neighbors are unchanged. (c) The objective score of the fixed ChemOPD grouping compared with 10,000 matched random assignments for each draw. Light and dark bars show the 1st–99th percentile range and interquartile range of the random scores. Diamonds show the fixed ChemOPD grouping.

Results. Figure 8(a) shows strong agreement between the resampled affinity matrices and the reference. The median Spearman correlation is 0.991, with a range of 0.981–0.994. The median Pearson correlation is 0.994, with a range of 0.989–0.995. Thus, both the ordering and the values of the 190 task-pair compatibilities remain stable. The median value of $\bar{J}_3^{(b)}$ is 0.950, with a range of 0.835–1.000. Figure 8(b) shows that most tasks keep the same three highest-scoring neighbors. A small number of tasks change one or more neighbors in some draws. However, all 48 high-magnitude reference pairs keep the same sign in every draw. The global affinity structure is therefore stable, although a few local rankings change. These local changes do not alter the selected task groups. All ten MIP runs recover the reference 8/8/6 grouping up to a permutation of teacher labels. LogP and solubility remain the two overlapping tasks in every run. Therefore, $J_{\text{group}}^{(b)} = 1$ and $\Delta F_b = 0$ for all ten draws. Figure 8(c) compares the fixed ChemOPD grouping with matched random assignments. The fixed grouping scores above all 10,000 random assignments in every draw. Its median standardized score is 9.39, with a range of 9.23–9.52. The selected grouping therefore remains well aligned with each resampled affinity matrix.

This experiment gives two results. First, the estimated task affinities are stable across different probe samples. Second, the MIP selects the same task groups in all ten runs, and these groups score far above matched random assignments. Together, the teacher-performance results in Section 5.2 and this resampling analysis provide evidence that affinity-guided grouping yields task-specific teacher gains and remains stable when the probe examples change.

G CHEMOPD ABLATIONS

This section provides additional ablation analysis for ChemOPD. We compare the full method with variants that remove the anchor or the warm-up schedule.

Anchor and specialist-weight schedule. Table 11 compares full ChemOPD with Specialist-only (MOPD) and no warm-up under the same initialization, specialists, routes, prompts, and update budget. Specialist-only removes the anchor, while no warm-up sets $\alpha_g(s) = 0.50$ from the first update. Full ChemOPD instead increases it from 0 to 0.50 during the first 65% of training. Against Specialist-only, full ChemOPD records 22 wins, four ties, and five losses across 31 metrics, includ-

ing an increase in Solubility property gain from 0.506 to 0.718. Against no warm-up, it records 23 wins, one tie, and seven losses. The main benefit appears in reaction reasoning, with nine wins among ten non-tied comparisons and an increase in Retrosynthesis FTS from 0.303 to 0.443. Together, these comparisons support combining the generalist anchor with a gradual specialist schedule: full ChemOPD improves more reported metrics than either specialist-only supervision or the no-warm-up variant.

Table 11: Component ablations for ChemOPD. Each panel follows the native metrics of Table 1. FG and Ring are lower-is-better.

(a) Molecule editing & understanding												
Method	EDITING			UNDERSTANDING								
	Add P@1	Delete P@1	Sub. P@1	Murcko Sim.	Equiv. Acc.	FG MAE ↓	Ring MAE ↓	Ring-Sys Sim.				
ChemOPD, Specialist-only	0.50	0.20	0.52	0.48	0.61	0.14	0.25	0.77				
ChemOPD, no warm-up	0.45	0.85	0.60	0.50	0.57	0.09	0.20	0.73				
ChemOPD	0.35	0.75	0.70	0.58	0.62	0.10	0.15	0.83				

(b) Molecular optimization												
Method	QED		LOGP		SOLUBILITY		DRD2		JNK3		GSK3B	
	SR	Δ	SR	Δ	SR	Δ	SR	Δ	SR	Δ	SR	Δ
ChemOPD, Specialist-only	0.76	0.167	0.70	0.641	0.63	0.506	0.50	0.032	0.18	-0.032	0.42	0.020
ChemOPD, no warm-up	0.84	0.191	0.63	0.500	0.61	0.427	0.45	0.055	0.32	-0.011	0.44	0.016
ChemOPD	0.86	0.167	0.68	0.601	0.71	0.718	0.55	0.098	0.29	-0.020	0.46	0.029

(c) Reaction reasoning												
Method	Fwd _{by}		Fwd _{major}		MECHSEL		NEPP		CONDITION		RETRO	
	FTS↑	Top-1	FTS↑	Top-1	Acc.	FTS↑	Top-1	FTS↑	Top-1	FTS↑	Top-1	
ChemOPD, Specialist-only	0.033	0.00	0.398	0.17	0.45	0.462	0.13	0.028	0.03	0.437	0.01	
ChemOPD, no warm-up	0.034	0.00	0.396	0.15	0.40	0.505	0.11	0.000	0.00	0.303	0.00	
ChemOPD	0.051	0.00	0.465	0.18	0.39	0.521	0.13	0.023	0.03	0.443	0.03	

H HEADROOM RECOVERY AND CAPABILITY DEVELOPMENT

H.1 HEADROOM RECOVERY

Headroom recovery relates the student improvements in Section 5.3 to the advantage offered by each routed specialist. Let j index the reported metric columns and c_j the associated subtask. Define the direction-aligned utility $U_j(\pi) = s_j m_j(\pi)$, where $m_j(\pi)$ is the reported score, $s_j = -1$ for MAE, and $s_j = 1$ otherwise. For student method a , routed-teacher headroom H_j , student gain $G_j^{(a)}$, and normalized recovery $\text{HR}_j^{(a)}$ are

$$H_j = U_j(T_{r(c_j)}) - U_j(\pi_{\theta_0}), \quad G_j^{(a)} = U_j(\pi_{\theta}^{(a)}) - U_j(\pi_{\theta_0}), \quad \text{HR}_j^{(a)} = \frac{G_j^{(a)}}{H_j}. \quad (19)$$

A negative recovery indicates regression from the common all-task SFT initializer. A value of one matches the routed teacher’s observed gain, and a value above one exceeds it. Figure 4(a–b) retains every reported metric column. The ratio summary uses the 26 columns with $H_j > 0.02$ to avoid near-zero denominators.

Positive recovery occurs on 25 of these columns for ChemOPD and 20 for MOPD. The reported medians are 0.62 and 0.37, respectively. Family medians for MOPD and ChemOPD are 0.11 and 0.51 for editing/understanding, 0.28 and 0.64 for optimization, and 0.53 and 0.68 for reaction. These comparisons keep the specialist checkpoints and routes fixed, so a stronger teacher collection cannot

explain the difference. Recovery is nonetheless a descriptive normalization of student improvement, not a causal measure of how much specialist-specific knowledge was transferred. Metric columns are not independent repeated trials.

H.2 CAPABILITY TRAJECTORIES AND TEACHER-RELATIVE SIGNALS

Capability summaries. Figure 9 connects capability development with the teacher signals observed during OPD. Let a index an OPD method, $s \in \{0, 27, 55, 82, 109\}$ denote the number of completed actor updates, and $\pi_{\theta_s}^{(a)}$ denote the corresponding student. The state at $s = 0$ is the common all-task SFT initialization. For summary curve r , let $\mathcal{J}_r$ contain its task metrics. We report the equal-weight mean

$$\overline{M}_r^{(a)}(s) = \frac{1}{|\mathcal{J}_r|} \sum_{j \in \mathcal{J}_r} m_j \left(\pi_{\theta_s}^{(a)} \right), \quad (20)$$

where $m_j(\pi)$ is the reported value of metric j for model π . Panels (a–c) show six summaries: editing P@1 over three tasks; understanding similarity or accuracy over three tasks; optimization success rate and property change over six properties; and reaction FTS and Top-1 over five molecular-output tasks. The understanding summary excludes the two count-error metrics, while the reaction summary excludes mechanism selection. Because the optimization property changes use different native units, their mean is used only to summarize the training trajectory.

Teacher-relative signals. For update $s \in \{1, \dots, 109\}$, let i index rollout sequences, ℓ index response positions, and $m_{sil} \in \{0, 1\}$ denote the valid-response mask. Let p_{sil}^S and $p_{sil}^{T_{c,i}}$ be the full-vocabulary student and teacher distributions at temperature $\tau = 0.7$. The channel $c \in \{A, T\}$ denotes the anchor or selected specialist, with $T_{A,i} = T_A$ and $T_{T,i} = T_{g(i)}$. Let $v_{sil k}$ be the token at rank k in the student top- K support. Its normalized student weight is

$$q_{sil k}^S = \frac{p_{sil}^S(v_{sil k})}{\sum_{j=1}^K p_{sil}^S(v_{sil j})}. \quad (21)$$

With $N_s = \sum_{i,\ell} m_{sil}$, the channel-wise discrepancy is

$$D_{s,c}^{(K)} = \frac{1}{N_s} \sum_{i,\ell} m_{sil} \sum_{k=1}^K q_{sil k}^S \left[\log p_{sil}^S(v_{sil k}) - \log p_{sil}^{T_{c,i}}(v_{sil k}) \right], \quad (22)$$

where $K = 16$. Panel (d) reports the mean over each non-overlapping 10-step bin:

$$\mathcal{B}_b = \{10(b-1) + 1, \dots, \min(10b, 109)\}, \quad \overline{D}_{b,c}^{(16)} = \frac{1}{|\mathcal{B}_b|} \sum_{s \in \mathcal{B}_b} D_{s,c}^{(16)}. \quad (23)$$

To interpret this quantity, consider one response position and define

$$Z^S = \sum_{k=1}^K p^S(v_k), \quad Z^{T_c} = \sum_{k=1}^K p^{T_c}(v_k), \quad q_k^{T_c} = \frac{p^{T_c}(v_k)}{Z^{T_c}}. \quad (24)$$

The token-level discrepancy satisfies

$$\sum_{k=1}^K q_k^S \log \frac{p^S(v_k)}{p^{T_c}(v_k)} = D_{\text{KL}}(q^S \| q^{T_c}) + \log \frac{Z^S}{Z^{T_c}}. \quad (25)$$

The support-mass term can be positive or negative. Therefore, $D_{s,c}^{(16)}$ is a signed top- K discrepancy rather than a metric distance.

ChemOPD improves all six summaries from the common initialization and achieves the highest final value on every curve. The comparison with MOPD is especially informative because both methods use the same initialization, specialists, routes, prompt stream, and update budget. At update 109, ChemOPD improves the editing and understanding summaries from 0.407 and 0.620 to 0.600 and 0.677 relative to MOPD. Optimization SR/ Δ increase from 0.532/0.222 to 0.592/0.266, while reaction FTS/Top-1 increase from 0.272/0.068 to 0.301/0.074. MOPD improves the reaction

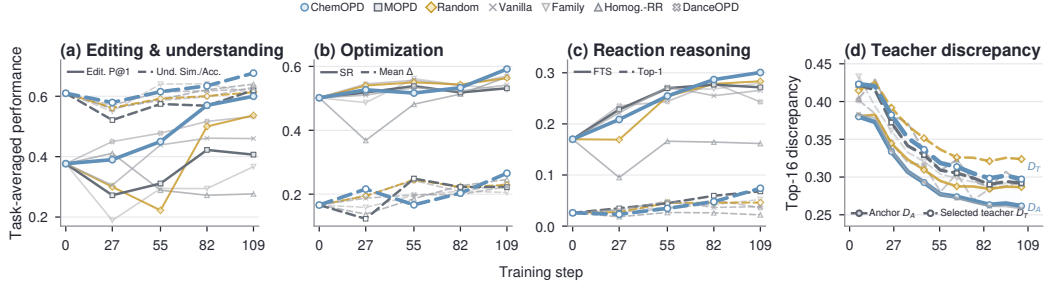


Figure 9: Capability trajectories and teacher-relative signals during OPD training. Panels (a–c) report the six task-averaged summaries defined in Equation 20. Line style distinguishes the two summaries in each panel. Panel (d) reports 10-step means of the anchor and selected-specialist discrepancies in Equation 22. The discrepancy curves are computed on each method’s own on-policy prefixes.

summaries more quickly early in training, whereas ChemOPD maintains a higher understanding summary and finishes above MOPD on both reaction summaries. This progression matches the intended schedule of retaining shared supervision while gradually introducing specialist guidance. The ChemOPD anchor discrepancy decreases from 0.380 to 0.262, and its specialist discrepancy decreases from 0.423 to 0.297. Across all bins, these curves differ from their Vanilla OPD and MOPD counterparts by at most 0.006 and 0.011, respectively. ChemOPD therefore retains both teacher-relative signal profiles while producing a stronger and more balanced capability profile. These results support scheduled anchor-residual supervision as an effective way to integrate heterogeneous specialist guidance into one student.

I OPD TRAINING DYNAMICS AND DIAGNOSTICS

This section examines how scheduled anchor-residual supervision changes the teacher-derived training signal and the resulting actor updates. The main comparison is ChemOPD versus MOPD. They use the same student initialization, specialists, routes, prompt stream, optimizer, and update budget. MOPD uses the routed specialist directly, whereas ChemOPD combines it with the all-task anchor using the schedule in Equation 7. This comparison therefore tests how the supervision rule affects specialist integration. The remaining methods provide additional references for the scale of the training signals and updates.

Training signal and actor update. We use the student top- K support and normalized weights defined in Equation 21, with $K = 16$. Let T_i denote the teacher applied to rollout sequence i , and let sg denote stop-gradient. The detached candidate-level advantage is

$$\hat{A}_{\text{sil}k}^{(T_i)} = -\text{sg}(q_{\text{sil}k}^S) \text{sg} \left[\log p_{\text{sil}k}^S(v_{\text{sil}k}) - \log p_{\text{sil}k}^{T_i}(v_{\text{sil}k}) \right]. \quad (26)$$

MOPD sets $T_i = T_{g(i)}$. ChemOPD computes the anchor and routed-specialist advantages separately and combines them according to Equation 6. The anchor supplies shared supervision, while the scheduled residual adds task-specific corrections.

Let $\ell_{\text{sil}k}^\theta$ be the current-student log-probability of candidate $v_{\text{sil}k}$ and let $\ell_{\text{sil}k}^{\text{old}} = \text{sg}(\ell_{\text{sil}k}^\theta)$. The actor probability ratio is

$$r_{\text{sil}k} = \exp \left[\text{clip}(\ell_{\text{sil}k}^\theta - \ell_{\text{sil}k}^{\text{old}}, -20, 20) \right]. \quad (27)$$

With PPO threshold $\epsilon = 0.2$ and dual-clip constant $\kappa = 3$, the loss term for each candidate is

$$u(\hat{A}, r) = \max \left\{ -\hat{A}r, -\hat{A} \text{clip}(r, 1 - \epsilon, 1 + \epsilon) \right\}, \quad (28)$$

$$\psi(\hat{A}, r) = \begin{cases} \min\{-\kappa\hat{A}, u(\hat{A}, r)\}, & \hat{A} < 0, \\ u(\hat{A}, r), & \hat{A} \geq 0. \end{cases} \quad (29)$$

The actor loss sums ψ over the K candidates and takes a masked mean over valid response positions. Losses from packed microbatches are weighted by their numbers of sequences before gradient accumulation. Each reported run uses one actor epoch and one logical minibatch per update. Therefore, $r = 1$ when an update is formed, and the local gradient is $-\hat{A}\nabla_{\theta} \log p_{\theta}$. No external correctness reward or additional KL penalty is used.

Training diagnostics. We use the update, response-mask, and 10-step-bin notation introduced in Appendix H.2. Let f_i denote the benchmark family of sequence i . For model $q \in \{S, T\}$, its response-token-weighted entropy for family f at update s is

$$\overline{\mathcal{H}}_{s,f}^q = \frac{\sum_{i:f_i=f} \sum_{\ell} m_{sil} \mathcal{H}(p_{sil}^q)}{\sum_{i:f_i=f} \sum_{\ell} m_{sil}}, \quad \mathcal{H}(p) = - \sum_{v \in \mathcal{V}} p(v) \log p(v). \quad (30)$$

Panel (a) of Figure 5 reports

$$E_b = \frac{1}{|\mathcal{F}_b|} \sum_{f \in \mathcal{F}_b} \frac{1}{|\mathcal{B}_{b,f}|} \sum_{s \in \mathcal{B}_{b,f}} \left| \overline{\mathcal{H}}_{s,f}^S - \overline{\mathcal{H}}_{s,f}^T \right|, \quad (31)$$

where $\mathcal{F}_b$ contains the families observed in bin b , and $\mathcal{B}_{b,f} \subseteq \mathcal{B}_b$ contains their updates. This macro-average gives each observed family equal weight. For ChemOPD, T is the routed MIP specialist rather than the anchor-specialist combination.

Panel (b) reports the signed aggregate top- K signal

$$S_s^{(K)} = -\frac{1}{N_s} \sum_{i,\ell} m_{sil} \sum_{k=1}^K \hat{A}_{sil k}. \quad (32)$$

For ChemOPD, $\hat{A}$ is the combined anchor-residual advantage. The negative sign follows the plotting convention. This quantity measures the signal passed to the actor; it is neither a gradient norm nor a full-vocabulary KL divergence. Panel (c) reports the percentage of valid response-position/candidate pairs with large advantages:

$$R_s^{\text{ext}} = 100 \frac{\sum_{i,\ell} m_{sil} \sum_{k=1}^K \mathbf{1}[\hat{A}_{sil k} > 5]}{K N_s}. \quad (33)$$

Panels (a–c) average these quantities within each $\mathcal{B}_b$. The relative reductions annotated in the figure are computed from all 109 update-level values before binning.

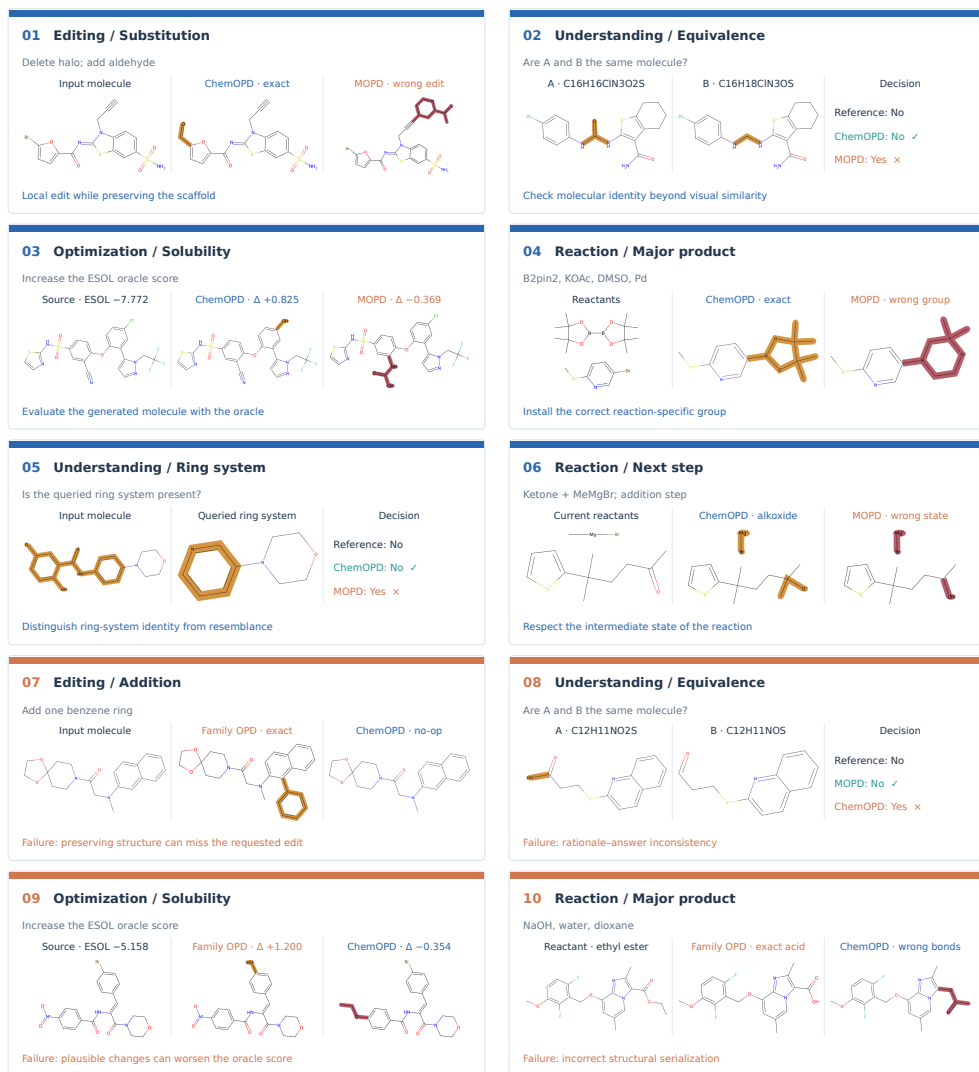
Let G_s be the total actor-gradient ℓ_2 norm measured before clipping. Panel (d) shows its empirical CDF and its post-warm-up exceedance rate:

$$\hat{F}_G(x) = \frac{1}{109} \sum_{s=1}^{109} \mathbf{1}[G_s \leq x], \quad C_{\text{post}} = \frac{1}{103} \sum_{s=7}^{109} \mathbf{1}[G_s > 1]. \quad (34)$$

The second statistic excludes the first six learning-rate warm-up updates. The value 1 is the shared gradient-clipping threshold.

ChemOPD has a mean student–specialist entropy gap of 0.0280, compared with 0.0313 for MOPD. This is a reduction of 10.6% and shows closer agreement in predictive sharpness. The aggregate top- K signal shows the effect of the schedule more directly. In the first 10-step bin, the ChemOPD signal is 0.381, close to anchor-only Vanilla OPD at 0.383 and below specialist-only MOPD at 0.422. As $\alpha_g(s)$ increases from 0 to 0.5, the ChemOPD curve moves toward the MOPD curve but remains lower in all 11 bins. Across all updates, the mean signal is 0.313 for ChemOPD and 0.334 for MOPD, a reduction of 6.2%.

The extreme-candidate rate decreases from 0.0314% for MOPD to 0.0222% for ChemOPD. This 29.4% reduction is observed in every bin. After learning-rate warm-up, 36.9% of ChemOPD updates exceed the gradient-clipping threshold, compared with 47.6% for MOPD and 35.9% for Vanilla OPD. ChemOPD therefore begins with an anchor-like update profile and introduces specialist corrections without reaching the larger signal tails of specialist-only training. Together with the capability gains in Section 5.3, these results show that the stronger ChemOPD student is not obtained by simply increasing the correction scale. Instead, the scheduled anchor-residual objective integrates specialist guidance through more moderate updates while retaining broad chemical capability.



Cases 1-6: ChemOPD succeeds against MOPD. Cases 7-10: comparator succeeds. Highlights follow the original chemical depictions.

Figure 10: Representative test-set examples across the four chemical task families. Cases 1–6, marked with blue headers, show examples in which ChemOPD produces the better evaluated result than MOPD. Cases 7–10, marked with orange headers, show examples in which the comparison method performs better; Case 8 uses MOPD, while the other three use benchmark-family OPD. Orange highlights indicate reference-consistent structural features, and rose highlights indicate incorrect structural changes. The reported ESOL scores are calculated by the same property evaluator used for benchmark scoring.

J CHEMISTRY-LEVEL CASE STUDIES

Figure 10 presents ten representative examples drawn from the held-out test sets. They cover a broad range of tasks across all four task families and illustrate the quality of the capabilities recovered by the unified student. The cases were selected for task coverage and chemical interpretability. Among these examples, ChemOPD produces the stronger task-level result in six cases, while the comparison method performs better in the remaining four. We examine whether the final outputs satisfy the task-specific chemical constraints and use representative responses to illustrate the types of capabilities evaluated by the benchmark.

Complete prompts and generated responses. For each case below, the input is shared because the two checkpoints were evaluated with the same prompt. The response blocks reproduce every generated output and the final answer.

Case 1: Local molecular substitution. ChemOPD returns the exact reference molecule and passes the substitution check. MOPD returns a valid molecule but retains the halo group and fails the requested edit.

Complete stored evaluation prompt (shared).

```
"You are a chemical assistant. Given the SMILES structural formula of a molecule, help me ADD and DELETE specified functional groups and output the modified SMILES sequence of the molecule. Input: Molecule SMILES string, Functional Group Names. Output: Modified Molecule SMILES string.
Your response must be directly parsable JSON format:
{
  "output": "Modified Molecule SMILES"
}
Input Molecule: C#CCn1/c(=N/C(=O)c2ccc(Br)o2)sc2cc(S(N)(=O)=O)ccc21,
Functional Group to delete: halo, Functional Group to add: aldehyde"
```

Complete ChemOPD generated response.

```
{
  "input_structure": "C#CCn1/c(=N/C(=O)c2ccc(Br)o2)sc2cc(S(N)(=O)=O)ccc21",
  "delete_analysis": "The input molecule contains a bromine atom (halo group) attached to a benzene ring in the substituent 'c2ccc(Br)o2'.",
  "add_analysis": "An aldehyde group (-CHO) can be added to the benzene ring at the position previously occupied by the bromine atom, converting it to a benzaldehyde (C=O).",
  "modification": "Replace the bromine atom (Br) with a formyl group (-CHO) on the benzene ring.",
  "output": "C#CCn1/c(=N/C(=O)c2ccc(C=O)o2)sc2cc(S(N)(=O)=O)ccc21"
}
```

Complete MOPD generated response.

```
{
  "input_structure": "C#CCn1/c(=N/C(=O)c2ccc(Br)o2)sc2cc(S(N)(=O)=O)ccc21",
  "delete_strategy": "Remove the bromine atom (halo group) from the aromatic ring at position 4 of the benzene ring.",
  "add_strategy": "Introduce an aldehyde group (-CHO) at the terminal position of the propyl chain attached to the nitrogen atom.",
  "output": "CC(=O)ClCC(C#CCn2/c(=N/C(=O)c3ccc(Br)o3)sc3cc(S(N)(=O)=O)ccc32)ccn1"
}
```

Case 2: Molecular equivalence. The reference label is “No.” ChemOPD returns “No” and is scored correct; MOPD returns “Yes” and is scored incorrect.

Complete stored evaluation prompt (shared).

```
You are a chemical assistant. Given two molecule SMILES, Please Determine whether these two Molecules are the same. Input: Molecule A SMILES string, Molecule B SMILES string. Output: yes / no.
Your response must be directly parsable JSON format:
{
  "output": "Yes / No"
}
```

Molecule A: NC(=O)c1c(NC(=O)Nc2ccc(Cl)cc2)sc2c1CCCC2, Molecule B: c12c(sc(NCNC3ccc(cc3)Cl)c2C(=O)N)CCCC1.

Complete ChemOPD generated response.

```
{
  "input_structure": "Molecule A: NC(=O)c1c(NC(=O)Nc2ccc(Cl)cc2)sc2c1CCCC2",
  "molecule_A_analysis": "A bicyclic system with a thiazole ring fused to a cyclohexane, containing an amide group (NC=O) attached to a benzene ring with a chlorine substituent and another amide group (NC=O) linked to the benzene.",
  "molecule_B_analysis": "A bicyclic system with a thiazole ring fused to a cyclohexane, containing an amide group (C(=O)N) attached to a benzene ring with a chlorine substituent and another amide group (C(=O)N) linked to the benzene.",
  "output": "No"
}
```

Complete MOPD generated response.

```
{
  "input_structure_A": "NC(=O)c1c(NC(=O)Nc2ccc(Cl)cc2)sc2c1CCCC2",
  "molecule_A_analysis": "Contains a thiazole ring fused to a cyclohexane, with two amide groups (NC=O) attached to the thiazole and a chlorophenyl group (c2ccc(Cl)cc2) via an amide linkage.",
  "input_structure_B": "c12c(sc(NCNC3ccc(cc3)Cl)c2C(=O)N)CCCC1",
  "molecule_B_analysis": "Contains a thiazole ring fused to a cyclohexane, with an amide group (C(=O)N) attached to the thiazole and a chlorophenyl group (c3ccc(cc3)Cl) via an amide linkage. The amide is connected to a secondary amine (NCN) instead of a primary amine.",
  "atom_count_comparison": "Both have 14 C, 3 N, 2 O, and 1 Cl. Atom types and counts match.",
  "functional_group_comparison": "Both have thiazole, amide, and chlorophenyl groups. The connectivity of the amide and chlorophenyl groups is identical, differing only in SMILES permutation (amide position and secondary amine substitution).",
  "output": "Yes"
}
```

Case 3: Solubility optimization. Both outputs are valid and retain the source Murcko scaffold. The ESOL deltas are +0.824958 for ChemOPD and -0.368879 for MOPD.

Complete stored evaluation prompt (shared).

You are a chemical assistant, Optimize the Source Molecule to improve the compound's ability to dissolve in water (Solubility) while following a structured intermediate optimization process.
Your response must be directly parsable JSON format:

```
{
  "Final Target Molecule": "SMILES",
}
```

Source Molecule: N#Cc1cc(S(=O)(=O)Nc2nccs2)ccc1Oc1ccc(Cl)cc1-c1ccnn1CC(F)(F)F.

Complete ChemOPD generated response.

```
{
  "Structural Analysis of Source Molecule": "The source molecule contains a nitrile group (N#C), a sulfonamide (S(=O)(=O)N), a thiazole ring (c2nccs2), a chlorophenyl group (Oc1ccc(Cl)cc1), and a trifluoromethyl-substituted pyridine (c1ccnn1CC(F)(F)F). The presence of the nitrile
```

```

    and sulfonamide groups may contribute to solubility, but the
    hydrophobic regions (e.g., chlorophenyl, trifluoromethyl) reduce
    water solubility.",
    "Property Analysis": "The molecule has moderate solubility due to polar
    groups (sulfonamide, nitrile), but the hydrophobic regions (
    chlorophenyl, trifluoromethyl) limit its water solubility. The
    thiazole ring (c2nccs2) is moderately polar but may not significantly
    enhance solubility.",
    "Limitation in Source Molecule for Property": "The chlorophenyl group (
    Oc1ccc(Cl)cc1) is highly hydrophobic, and the trifluoromethyl-
    substituted pyridine (c1ccnn1CC(F)(F)F) adds to the hydrophobicity.
    The nitrile group (N#C) is polar but may not compensate enough for
    the hydrophobic regions.",
    "Optimization for Source Molecule": "Replace the chlorophenyl group (
    Oc1ccc(Cl)cc1) with a more polar substituent, such as a hydroxyl
    group (O), to improve water solubility. The trifluoromethyl-
    substituted pyridine (c1ccnn1CC(F)(F)F) can be retained as it may
    contribute to activity, but the hydrophobicity of the pyridine ring
    can be mitigated by introducing additional polar groups elsewhere in
    the molecule.",
    "Final Target Molecule": "N#Cc1cc(S(=O)(=O)Nc2nccs2)ccc1Oc1ccc(O)cc1-
    c1ccnn1CC(F)(F)F"
}

```

Complete MOPD generated response.

```

{
  "Structural Analysis of Source Molecule": "The source molecule contains
  a nitrile group (N#C), a sulfonamide group (S(=O)(=O)N), a thiazole
  ring (c2nccs2), a chlorophenyl group (Oc1ccc(Cl)cc1), and a
  trifluoromethyl-substituted pyridine (c1ccnn1CC(F)(F)F). The presence
  of the nitrile and sulfonamide groups may contribute to solubility,
  but the aromatic systems and halogen substituents reduce water
  solubility.",
  "Property Analysis": "The molecule has moderate solubility due to the
  sulfonamide and nitrile groups, but the aromatic rings and halogens (
  Cl, F) limit its water solubility. The trifluoromethyl group (CC(F)(F)
  )F) is hydrophobic and further reduces solubility.",
  "Limitation in Source Molecule for Property": "The main limitations are
  the hydrophobic aromatic systems (chlorophenyl, pyridine with
  trifluoromethyl), the nitrile group (which is polar but may not fully
  compensate for hydrophobicity), and the lack of additional polar or
  ionizable groups.",
  "Optimization for Source Molecule": "To improve solubility, we can: 1)
  Replace the nitrile with a more polar group (e.g., hydroxyl or amine)
  , 2) Introduce a polar substituent (e.g., hydroxyl or amine) on the
  aromatic rings, 3) Replace the trifluoromethyl group with a more
  polar group (e.g., hydroxyl or amine), and 4) Add a hydrophilic group
  (e.g., carboxylic acid or amine) to the pyridine ring.",
  "Final Target Molecule": "O=C(O)Nc1cc(S(=O)(=O)Nc2nccs2)ccc1Oc1ccc(Cl)
  cc1-c1ccnn1CC(F)(F)F"
}

```

Case 4: Miyaura borylation product. ChemOPD's final molecule exactly matches the reference, whereas MOPD's valid molecule does not. ChemOPD's reaction-class and solvent descriptions remain incorrect, as retained below.

Complete stored evaluation prompt (shared).

Given the reactants SMILES and reagents SMILES, your task is to predict the main product SMILES using your experienced chemical Reaction Prediction knowledge.

```

Input:
Reactants SMILES separated by '.'
Reagents SMILES separated by '.'

Output:
Main product SMILES
[Optional: Byproduct SMILES separated by '.']

The output SMILES must be valid and chemically reasonable. Do not provide
any additional information beyond the requested SMILES strings. The
answer should be a json format that includes the main product SMILES
and byproduct SMILES if any:

Question:
Reactants: CC1(C)OB(B2OC(C)(C)C(C)(C)O2)OC1(C)C.CSclccc(Br)cn1
Reagents: CC(=O)O[K].CS(C)=O.[Pd]

Answer:
{
  "Major Product": ...
  "Byproduct(s)": ...
}

```

Complete ChemOPD generated response.

```

{
  "Parse Reactants/Reagents": "Reactant 1: CC1(C)OB(B2OC(C)(C)C(C)(C)O2)OC1(C)C (Bis(pinacolato)diboron, B2pin2). Reactant 2: CSclccc(Br)cn1 (2-bromo-5-(methylthio)pyridine). Reagents: CC(=O)O[K] (Potassium acetate), CS(C)=O (Sulfolane, a solvent).",
  "Identify Functional Groups": "Reactant 1: Diboron reagent (source of B-B bond). Reactant 2: Aryl bromide (C-Br bond on pyridine ring), methylthio group (-SMe), pyridine ring. Reagents: Potassium acetate (salt), Sulfolane (solvent).",
  "Predict Reaction Type": "Buchwald-Hartwig amination. This is a palladium-catalyzed cross-coupling reaction between an aryl halide (the bromo-pyridine derivative) and a boronic ester (bis(pinacolato)diboron) to form a C-N bond. The reaction is typically performed in the presence of a base (like potassium acetate) and a palladium catalyst (e.g., Pd(PPh3)4).",
  "Construct Product Structure": "The bromine atom on the pyridine ring of CSclccc(Br)cn1 is replaced by the 2-(pinacolato)diboron group from B2pin2. The pinacolato group (B(pin)2) is attached to the carbon atom of the pyridine ring that was previously bonded to the bromine. The methylthio group (-SMe) and the pyridine ring structure remain unchanged. The product is 2-(2-(pinacolato)diboron)-5-(methylthio)pyridine.",
  "Final_SMILES": "CSclccc(B2OC(C)(C)C(C)(C)O2)cn1"
}

```

Complete MOPD generated response.

```

{
  "Parse Reactants/Reagents": "Reactant 1: CC1(C)OB(B2OC(C)(C)C(C)(C)O2)OC1(C)C (Bis(pinacolato)diboron, B2pin2). Reactant 2: CSclccc(Br)cn1 (2-bromo-5-(methylthio)pyridine). Reagents: CC(=O)O[K] (Potassium acetate), CS(C)=O (Sulfonyl chloride), [Pd] (Palladium catalyst).",
  "Identify Functional Groups": "Reactant 1: Diboron reagent (source of B-B bond, B-B bond is the reactive site). Reactant 2: Aryl bromide (C-Br bond on pyridine ring), methylthio group (-SMe), pyridine ring. Reagents: Potassium acetate (salt), Sulfonyl chloride (source of -SO2Cl, which is a good leaving group), Palladium (catalyst).",
  "Predict Reaction Type": "Palladium-catalyzed Buchwald-Hartwig amination. This reaction couples an aryl halide (in this case, an

```

```

    aryl bromide) with an amine (in this case, a pyridine derivative) to
    form a new carbon-nitrogen bond. The aryl bromide is the electrophile
    , and the pyridine derivative is the nucleophile. The reaction is
    typically promoted by a palladium catalyst and a base (potassium
    acetate is used as a base).",
    "Construct Product Structure": "The reaction involves the formation of
    a new C-N bond between the pyridine ring of CSc1ccc(Br)cn1 and the
    carbon atom of the pyridine ring that was previously bonded to the
    bromine atom. The bromine atom is replaced by the pyridine nitrogen
    atom. The methylthio group (-SMe) and the pinacolboronate ester group
    (-Bpin) are expected to remain unchanged under these conditions. The
    product is 2-(methylthio)-5-(pinacolboronato)pyridine. The SMILES
    string for the product is CSc1ccc(N2CCOC(C)(C)C2)cn1.",
    "Final_SMILES": "CSc1ccc(N2CCOC(C)(C)C2)cn1"
  }

```

Case 5: Ring-system recognition. The reference label is “No.” ChemOPD returns “No” and is scored correct; MOPD returns “Yes” and is scored incorrect.

Complete stored evaluation prompt (shared).

```

You are a chemical assistant. Please Determine whether the
  ring_system_scaffold is in the Molecule. Input: a molecule's SMILES
  string, a Ring System Scaffold. Output: yes / no.
Definition: The ring system scaffold consists of one or more cyclic (ring
  -shaped) molecular structures.
Your response must be directly parsable JSON format:
{
  "output": "Yes / No"
}
Input Molecule: O=C(Nc1ccc(N2CCOCC2)cc1)c1cc(Cl)ccc1O, Ring System
  Structure: c1cncc(N2CCOCC2)c1

```

Complete ChemOPD generated response.

```

{
  "input_structure": "O=C(Nc1ccc(N2CCOCC2)cc1)c1cc(Cl)ccc1O",
  "molecule_structure_analysis": "Contains a benzamide group (amide
    attached to a benzene ring with a morpholine substituent and a
    hydroxyl group on a chlorobenzene ring).",
  "scaffold_analysis": "A pyridine ring (c1cncc) with a morpholine
    substituent (N2CCOCC2).",
  "matching_analysis": "The molecule includes a benzene ring (c1ccc...cc1
    ) with a morpholine substituent, but the scaffold's pyridine ring (
    c1cncc) is absent. The benzene ring is a non-aromatic cyclic
    structure, not matching the scaffold's pyridine core.",
  "output": "No"
}

```

Complete MOPD generated response.

```

{
  "input_structure": "O=C(Nc1ccc(N2CCOCC2)cc1)c1cc(Cl)ccc1O",
  "molecule_structure_analysis": "Contains a benzamide group linked to a
    4-chlorophenol and a 2,5-dihydro-1,3-dioxolane (morpholine)
    substituent.",
  "scaffold_analysis": "A pyridine ring (c1cncc) with a morpholine
    substituent (N2CCOCC2).",
  "matching_analysis": "The molecule includes a pyridine ring (benzamide's
    aromatic system) connected to morpholine, matching the scaffold's
    core structure.",
  "output": "Yes"
}

```


Case 6: Grignard next elementary step. ChemOPD emits the reference magnesium-alkoxide state. MOPD emits a protonated alcohol and omits the carbon introduced by methylmagnesium bromide.

Complete stored evaluation prompt (shared).

```
We have one typical reaction (reaction class: 'Bromo Grignard + ester
reaction', starting reactants are 'CCOC(=O)CCC(C)(C)ClCCCS1.C[Mg]Br',
reagents are 'CCOCC.[Cl-].C[Mg]Br.[NH4+].O', and reaction condition
is 'Reaction').

Here are the previous elementary reaction steps:

Elementary Step 1:

{
  "reactants": C[Mg]Br.CCOC(=O)CCC(C)(C)ClCCCS1,
  "products": CCOC(C)([O-])CCC(C)(C)ClCCCS1.[Mg+]Br,
  "step annotation": Addition,
}

Elementary Step 2:

{
  "reactants": CCOC(C)([O-])CCC(C)(C)ClCCCS1,
  "products": CC[O-].CC(=O)CCC(C)(C)ClCCCS1,
  "step annotation": Leaving or workup,
}

Now, we want to predict the next elementary reaction step. Currently we
know the basic information:
"current_step_info": {
  "reactants": C[Mg]Br.CC(=O)CCC(C)(C)ClCCCS1,
  "step annotation": Addition,
}

Under the same reaction condition and reagents, please give me the
products of the next step element reaction. Just return the SMILES of
prediction. Your response must contains directly parsable JSON
format:

{
  "pred_smi": str
}
```

Complete ChemOPD generated response.

```
Step-1, Identify Reaction Mechanism, The reaction is a Grignard reaction
with an ester. The Grignard reagent (C[Mg]Br) acts as a strong
nucleophile, attacking the electrophilic carbonyl carbon of the ester
. Step-2, Analyze Previous Steps, Step 1 showed the nucleophilic
addition of the Grignard reagent to the ester carbonyl, forming a
tetrahedral intermediate. Step 2 showed the collapse of this
intermediate, resulting in a carboxylate anion and a ketone. The
current step involves the same reactants, but the context is the
initial addition of the Grignard reagent to the ester. Step-3,
Determine Current Reactants Context, The reactants are the Grignard
reagent, C[Mg]Br, and the ester, CC(=O)CCC(C)(C)ClCCCS1. The Grignard
reagent is a strong nucleophile, and the ester is an electrophile.
The reaction is an addition of the Grignard reagent to the carbonyl
group. Step-4, Apply Mechanistic Logic, The nucleophilic carbon of
the Grignard reagent (C[Mg]Br) will attack the electrophilic carbonyl
```

carbon of the ester (CC(=O)CCC(C)(C)ClCCCS1). This is a classic nucleophilic addition to a carbonyl group. The pi-bond of the carbonyl will break, and the electrons will move to the oxygen atom, forming a new C-O bond. The magnesium bromide will remain as a counterion. Step-5, Construct Products, The attack of the Grignard reagent on the ester carbonyl carbon results in the formation of a new C-C bond. The original carbonyl group becomes a tertiary alcohol. The product is a magnesium alkoxide intermediate, specifically a magnesium alkoxide salt of the alcohol. The SMILES for this intermediate is CC(C)([O-])CCC(C)(C)ClCCCS1.[Mg+]Br. Step-6, Validate Reagents Conditions, The reaction is carried out in a Grignard reaction solvent (like THF or diethyl ether) and with a base (like ammonium chloride or a salt like magnesium bromide). The formation of a magnesium alkoxide intermediate is a standard and expected step in the Grignard addition to an ester. The conditions are consistent with this transformation. Output: `<answer> CC(C)([O-])CCC(C)(C)ClCCCS1.[Mg+]Br </answer>`

Complete MOPD generated response.

Step-1, Identify Reaction Mechanism, The reaction is a Grignard reaction with an ester. The first step was the nucleophilic addition of the Grignard reagent to the ester carbonyl, forming a tetrahedral intermediate. The current step involves the addition of the Grignard reagent to the newly formed ketone intermediate. Step-2, Analyze Previous Steps, Step 1 showed the addition of the Grignard reagent (C[Mg]Br) to the ester (CCOC(=O)CCC(C)(C)ClCCCS1), forming a tetrahedral intermediate (CCOC(C)([O-])CCC(C)(C)ClCCCS1) and a magnesium bromide cation ([Mg+]Br). Step 2 showed the collapse of this tetrahedral intermediate, resulting in a carboxylate anion (CC([O-])) and a ketone (CC(=O)CCC(C)(C)ClCCCS1). Step-3, Determine Current Reactants Context, The current reactants are the ketone intermediate (CC(=O)CCC(C)(C)ClCCCS1) and the Grignard reagent (C[Mg]Br). The ketone is electrophilic at the carbonyl carbon, and the Grignard reagent is a strong nucleophile. Step-4, Apply Mechanistic Logic, The nucleophilic carbanion of the Grignard reagent (C[Mg]Br) will attack the electrophilic carbonyl carbon of the ketone. This is a classic nucleophilic addition to a carbonyl group. The pi-bond of the carbonyl will break, and the electrons will move to the oxygen atom, forming a new C-O bond. Step-5, Construct Products, The attack of the Grignard reagent on the ketone carbonyl carbon results in the formation of a tertiary alcohol. The carbon atom of the original ketone becomes a tertiary alcohol, and the magnesium bromide cation ([Mg+]Br) is released. The product is the tertiary alcohol (CC(O)CCC(C)(C)ClCCCS1) and the magnesium bromide cation ([Mg+]Br). Step-6, Validate Reagents Conditions, The reaction is carried out in the presence of a Grignard reagent and a base (NH4+). The formation of a tertiary alcohol is a standard and expected outcome for a Grignard reaction with a ketone, especially if the reaction is not quenched. The magnesium bromide cation is a common byproduct. Output: `<answer> CC(O)CCC(C)(C)ClCCCS1.[Mg+]Br </answer>`

Case 7: Explicit ring addition. Benchmark-family OPD returns the exact reference, while ChemOPD copies the input molecule and fails the requested addition.

Complete stored evaluation prompt (shared).

You are a chemical assistant. Given the SMILES structural formula of a molecule, help me add a specified functional group and output the improved SMILES sequence of the molecule. Input: Molecule SMILES string, Functional Group Name. Output: Modified Molecule SMILES string.

Your response must be directly parsable JSON format:

```
{
```

```

    "output": "Modified Molecule SMILES"
  }

Input Molecule: CN(CC(=O)N1CCC2(CC1)OCCO2)c1ccc2ccccc2c1, Functional
Group to add: benzene_ring.

```

Complete ChemOPD generated response.

```

{
  "input_structure": "CN(CC(=O)N1CCC2(CC1)OCCO2)c1ccc2ccccc2c1",
  "functional_group": "benzene_ring",
  "output": "CN(CC(=O)N1CCC2(CC1)OCCO2)c1ccc2ccccc2c1"
}

```

Complete benchmark-family OPD generated response.

```

{
  "input_structure": "CN(CC(=O)N1CCC2(CC1)OCCO2)c1ccc2ccccc2c1",
  "functional_group_addition": "The benzene_ring is added as a
    substituent to the existing benzene ring in the molecule.",
  "add_strategy": "The benzene_ring is appended to the existing benzene
    ring at the position where the ethyl group (c1ccc2ccccc2c1) is
    attached, forming a biphenyl system.",
  "feasibility_analysis": "The modification is chemically viable as it
    involves attaching a benzene ring to an existing aromatic system
    without disrupting the core structure. The resulting biphenyl system
    is stable.",
  "output": "CN(CC(=O)N1CCC2(CC1)OCCO2)c1ccc2ccccc2c1c1ccccc1"
}

```

Case 8: Acid-aldehyde equivalence. The reference label is “No.” MOPD returns “No” and is scored correct; ChemOPD returns “Yes” and is scored incorrect.

Complete stored evaluation prompt (shared).

```

You are a chemical assistant. Given two molecule SMILES, Please Determine
whether these two Molecules are the same. Input: Molecule A SMILES
string, Molecule B SMILES string. Output: yes / no.
Your response must be directly parsable JSON format:
{
  "output": "Yes / No"
}
Molecule A: O=C(O)CCSc1ccc2ccccc2n1, Molecule B: C(=O)CCSc1ccc2ccccc2n1.

```

Complete ChemOPD generated response.

```

{
  "input_structure": "Molecule A: O=C(O)CCSc1ccc2ccccc2n1 (contains a
    carboxylic acid group, thioether, and a fused bicyclic aromatic
    system). Molecule B: C(=O)CCSc1ccc2ccccc2n1 (contains a carbonyl
    group, thioether, and the same fused bicyclic aromatic system).",
  "atom_count_analysis": "Both molecules have 15 carbons, 1 oxygen, 1
    sulfur, and 1 nitrogen. The carboxylic acid in A is replaced by a
    carbonyl in B, but the core structure (thioether + fused aromatic
    system) remains identical.",
  "functional_group_analysis": "Molecule A has a carboxylic acid (-COOH)
    group, while Molecule B has a carbonyl (-C=O) group. The functional
    group differences are due to the substitution of -OH with -C=O, but
    the connectivity of the thioether and aromatic system is preserved.",
  "output": "Yes"
}

```

Complete MOPD generated response.

```
{
  "input_structure_A": "O=C(O)CCSc1ccc2ccccc2n1",
  "molecule_A_analysis": "A carboxylic acid group (O=C(O)) attached to a propyl chain (CC) connected to a thioether (S) linking to a fused bicyclic aromatic system (benzene fused with a pyridine-like ring).",
  "input_structure_B": "C(=O)CCSc1ccc2ccccc2n1",
  "molecule_B_analysis": "A carboxylic acid group (C(=O)) attached to a propyl chain (CC) connected to a thioether (S) linking to a fused bicyclic aromatic system (benzene fused with a pyridine-like ring).",
  "atom_count_comparison": {
    "A": {
      "C": 10,
      "O": 2,
      "S": 1,
      "N": 1
    },
    "B": {
      "C": 10,
      "O": 1,
      "S": 1,
      "N": 1
    },
    "output": "No",
    "atom_count_mismatch": {
      "O": 1,
      "output": "No",
      "atom_count_difference": {
        "O": 1
      }
    }
  },
  "bond_count_comparison": {
    "A": 10,
    "B": 10,
    "output": "No",
    "bond_count_mismatch": {
      "output": "No",
      "bond_count_difference": {}
    }
  },
  "functional_group_comparison": {
    "A": "Carboxylic acid, thioether, fused bicyclic aromatic system",
    "B": "Carboxylic acid, thioether, fused bicyclic aromatic system",
    "output": "No",
    "functional_group_identicality": "Yes",
    "functional_group_mismatch": "Carboxylic acid group: O=C(O) vs C(=O) in SMILES representation"
  },
  "output": "No"
}
```

Case 9: Solubility-oracle boundary. The ESOL delta is +1.199869 for benchmark-family OPD and -0.354096 for ChemOPD; both outputs are valid and retain the source Murcko scaffold.

Complete stored evaluation prompt (shared).

```
You are a chemical assistant, Optimize the Source Molecule to improve the compound's ability to dissolve in water (Solubility) while following a structured intermediate optimization process.
Your response must be directly parsable JSON format:
{
  "Final Target Molecule": "SMILES",
```

```
}
Source Molecule: O=C(NC(=Cc1ccc(Br)cc1)C(=O)N1CCOCC1)c1ccc([N+](=O)[O-])cc1.
```

Complete ChemOPD generated response.

```
{
  "Structural Analysis of Source Molecule": "The source molecule contains a central amide group (O=C-N) connected to a brominated benzyl group (C(=Cc1ccc(Br)cc1)) and a nitro-substituted benzene ring (c1ccc([N+](=O)[O-])cc1). The nitro group ([N+](=O)[O-]) is highly polar but may limit solubility due to its strong electron-withdrawing nature.",
  "Property Analysis": "The nitro group contributes to high polarity but may reduce water solubility due to its strong electron-withdrawing effect, which can destabilize the molecule. The brominated benzyl group adds hydrophobicity.",
  "Limitation in Source Molecule for Property": "The nitro group ([N+](=O)[O-]) is a major limitation for water solubility due to its high polarity and potential for hydrogen bonding disruption. The bromine atom also adds hydrophobicity.",
  "Optimization for Source Molecule": "Replace the nitro group ([N+](=O)[O-]) with a methoxy group (OCH3) to reduce polarity and improve solubility. The methoxy group is less electron-withdrawing and more hydrophilic, enhancing water solubility while maintaining some aromatic character.",
  "Final Target Molecule": "O=C(NC(=Cc1ccc(Br)cc1)C(=O)N1CCOCC1)c1ccc(OCC)cc1"
}
```

Complete benchmark-family OPD generated response.

```
{
  "Structural Analysis of Source Molecule": "The source molecule contains a central amide group (O=C-N) connected to a brominated benzyl group (C(=Cc1ccc(Br)cc1)) and a nitro-substituted benzene ring (c1ccc([N+](=O)[O-])cc1). The nitro group ([N+](=O)[O-]) is highly polar and electron-withdrawing, while the bromine (Br) is a hydrophobic substituent.",
  "Property Analysis": "The nitro group enhances water solubility due to its polarity and ability to form hydrogen bonds. However, the bromine atom reduces solubility by introducing hydrophobicity. The amide and ether groups (C(=O)N1CCOCC1) contribute to moderate solubility but are insufficient to counteract the hydrophobic effects of Br and the nitro group.",
  "Limitation in Source Molecule for Property": "The bromine atom is the primary limitation for water solubility due to its hydrophobic nature. The nitro group, while polar, is not enough to offset the hydrophobicity introduced by Br.",
  "Optimization for Source Molecule": "Replace the bromine atom with a more polar and hydrophilic group, such as a hydroxyl (-OH) or a carboxyl (-COOH) group, to enhance water solubility. The nitro group can be retained to maintain polarity and hydrogen bonding capacity.",
  "Final Target Molecule": "O=C(NC(=Cc1ccc(O)cc1)C(=O)N1CCOCC1)c1ccc([N+](=O)[O-])cc1"
}
```

Case 10: Ester saponification. Benchmark-family OPD exactly matches the reference carboxylic acid. ChemOPD's product is valid but has the wrong connectivity and does not match the reference.

Complete stored evaluation prompt (shared).

Given the reactants SMILES and reagents SMILES, your task is to predict the main product SMILES using your experienced chemical Reaction Prediction knowledge.

Input:
 Reactants SMILES separated by '.'
 Reagents SMILES separated by '.'

Output:
 Main product SMILES
 [Optional: Byproduct SMILES separated by '.']

The output SMILES must be valid and chemically reasonable. Do not provide any additional information beyond the requested SMILES strings. The answer should be a json format that includes the main product SMILES and byproduct SMILES if any:

Question:
 Reactants: CCOC(=O)c1c(C)nc2c(OCc3c(F)ccc(OC)c3F)cc(C)cn12
 Reagents: O.O[Na].C1COCCO1

Answer:

```
{
  "Major Product": ...
  "Byproduct(s)": ...
}
```

Complete ChemOPD generated response.

```
{
  "Parse Reactants/Reagents": "Reactant 1: CCOC(=O)c1c(C)nc2c(OCc3c(F)ccc(OC)c3F)cc(C)cn12 (an ethyl ester). Reagent 1: O.O[Na].C1COCCO1 (1,4-Dioxane-2,5-diol, a cyclic ether).",
  "Identify Functional Groups": "Reactant 1: Ethyl ester (-COOEt), pyrazolo[1,5-a]pyrimidine core, methyl groups, methoxy groups on a substituted phenyl ring, and a fluorine atom. Reagent 1: 1,4-Dioxane-2,5-diol (a cyclic ether, specifically a 1,4-dioxane derivative, which can act as a nucleophile or a base).",
  "Predict Reaction Type": "The reaction is predicted to be a base-catalyzed ester hydrolysis (saponification). The 1,4-dioxane-2,5-diol (reagent 1) acts as a base, and the ethyl ester (reactant 1) is the substrate. The reaction proceeds via nucleophilic acyl substitution, where the oxygen of the 1,4-dioxane-2,5-diol attacks the carbonyl carbon of the ester, leading to the cleavage of the C-OEt bond and formation of a carboxylate salt and ethanol. Subsequent protonation (typically during aqueous workup) yields the carboxylic acid.",
  "Construct Product Structure": "The ethyl ester group (-C(=O)OCC) is hydrolyzed to a carboxylic acid group (-C(=O)OH). The rest of the molecule, including the pyrazolo[1,5-a]pyrimidine core, the methyl groups, the methoxy groups on the phenyl ring, and the fluorine atom, remains unchanged. The product is the carboxylic acid: CC(=O)Oc1c(C)nc2c(OCc3c(F)ccc(OC)c3F)cc(C)cn12.",
  "Final_SMILES": "CC(=O)Oc1c(C)nc2c(OCc3c(F)ccc(OC)c3F)cc(C)cn12"
}
```

Complete benchmark-family OPD generated response.

```
{
  "Parse Reactants/Reagents": "Reactant 1: CCOC(=O)c1c(C)nc2c(OCc3c(F)ccc(OC)c3F)cc(C)cn12 (an ethyl ester). Reagent 1: O (water). Reagent 2: O[Na] (sodium hydroxide). Reagent 3: C1COCCO1 (1,4-dioxane, a solvent).",
  "Identify Functional Groups": "Reactant 1 contains an ethyl ester group (-C(=O)OCC), a pyrazolo[1,5-a]pyrimidine core, a methoxy group (-
```

```

    OCH3), and a difluoromethoxy group (-OCH2F). The ester group is the
    most reactive site for the reaction with sodium hydroxide.",
    "Predict Reaction Type": "Saponification (base-catalyzed hydrolysis of
    an ester). The hydroxide ion (OH-) from sodium hydroxide attacks the
    electrophilic carbonyl carbon of the ester. This leads to the
    cleavage of the acyl-oxygen bond, forming a carboxylate salt and
    ethanol. Subsequent protonation (typically during aqueous workup)
    yields the carboxylic acid.",
    "Construct Product Structure": "The ethyl ester group (CCOC(=O)-) is
    converted into a carboxylic acid group (O=C(O)-). The rest of the
    molecule, including the pyrazolo[1,5-a]pyrimidine core, the methoxy
    group, and the difluoromethoxy group, remains unchanged. The SMILES
    fragment CCOC(=O)- becomes O=C(O)-.",
    "Final_SMILES": "O=C(O)c1c(C)nc2c(OCc3c(F)ccc(OC)c3F)cc(C)cn12"
  }

```